Can power-side monitoring see a 16-chip training island? An annotated outline

PROVISIONAL DRAFT, 2026-10-16.

Figures, framing and wording will change.

Distributed training is a problem for governance (Rahman 2026). Instead of one large cluster, which is easy to find by its heat, its construction and its power draw, a run can be spread across hundreds of small sites, which might allow covert training.

Here we examine training islands at the smallest scale proposed for governance (16 chips) and one step above (32 chips). Our best guess based upon the preliminary data and simulations is that 32-chip islands would be visible by default, and would require deliberate effort to hide; the number of locations that they COULD hide is large but definitely bounded. It almost certainly could do something like "narrow the range of possibilities where we need to take a closer look"; that might e.g. qualify for a warrant or similar followup. We find 16-chip islands are at the edge of detectability, but are optimistic that most would show up *sometimes* and/or be predicted based on ML etc.

Briefly, feeder lines come out of the distribution substations, typically carrying ~10 MW total, split across 3 phases (A, B, C) of ~3 MW each. A 16-chip island is ~27 kW, call it 30 kW, i.e. 1% of one phase. Power monitors are already installed widely throughout the grid, and ±1% for current is pretty common, so the smallest islands are at the edge of detection in absolute terms. But the *dynamic* accuracy of these monitors is typically better, so changes in the signal are much easier to detect: when an island turns on or off, and the pulse it makes with every training step, roughly every ten seconds.

This is made simpler because while there are plenty of motors that work at this power level, the onset behavior for a motor is qualitatively different.[1] Onset typically peaks at several times the steady-state power, ramps up as the motor starts to spin, and introduces a phase angle in the power system (aka "reactive load"). An island starts almost instantly with no phase angle; closer in nature to a water heater, but most places don't run 30 kW water heaters for 20+ hours a day.

To test this, we took real datasets from 2 different grids (Caltech and LBNL) and a real dataset of power consumption during model training (from NREL), and added the model-training data to the grid data at levels from a single building up to a campus line about the size of a feeder. In some cases, this is detectable with the "binocular statistic" (you can see the difference with your own two eyes). Some take a bit more analysis, but we haven't even scratched the surface of more sophisticated time-series modeling and anomaly detection. There are ways an evader might try to hide it, obviously, but some of those become detectable themselves. The cheapest is to spread the island evenly across three phases, which takes deliberate cabling.

The primary caveat right now is that we haven't actually installed an electric monitor before the meter to check; this is real grid data and real training data, added together. All of this needs experimental, hands-on testing, and we hope to take measurements in both the US (the background for our simulations) and the UK (which has several important differences).

[1] Motors behind variable-speed drives start gently, without the phase angle, and look more like an island (section 2.4).

Overall approach. We take current and power records from meters on real, working distribution grids (a 12 kV campus bus at LBNL, read 120 times a second; three meters on the Caltech campus, from a 252 kVA building main to a 5.5 MVA campus supply line, plus an energy balance across one switchgear section, read once a second) and treat them as the background a monitor would face in the field. To that background we add, by arithmetic, GPU power readings from a real 16-H100 training job (a Llama-2 70B LoRA fine-tune; a Stable Diffusion run serves as a check). We then compare the record with and without the training signal and ask under which conditions a detector can tell them apart: at which aggregation level, on which phase arrangement, with which signal, and at what false-alarm cost. For the pulse results the grid record and the job's GPU power readings are both real. The job trace is cut to its busy stretch, mean-removed, resampled, looped to fill each hour and added (kW taken as kVA). The step results count natural jumps, and the one injected step test is synthetic. The main limits are that the job traces come from the GPUs' own power counters rather than from the wall, and that the headline rests on three logged jobs from two labs, the third of which falls short at the largest meter at margin 2 (sections 2.3 and 2.8).

How to read the tags. Every figure carries one tag.

- **[M]** Measured. A quantity read from recorded data.
- **[S-job]** Simulated with a logged job. A measured training-job power trace is added to a measured grid record by arithmetic.
- **[S-syn]** Simulated with a synthetic signal. A made-up waveform is added to a measured record.
- **[A-src]** Asserted. Someone on this project read the primary source.
- **[A-chk]** Asserted. A later source check confirmed the figure in the primary source.
- **[A-rep]** Asserted. The figure comes from an unverified research report or summary. Nobody has checked the primary source. Appendix C lists these.
- **[A-arith]** Asserted. Agent arithmetic or an assumption.
- **[S-sim]** Simulated with a cost model. Output of the companion outline's runs of Rahman's distributed-training model (*What does a 16-chip cluster limit cost an evader?*), which gives the island counts used here.
- **[no source found]** Added after a tag where the 2026-10-09 source check found no source.

Terms. The project's names, with the standard name where one exists.

Term	Meaning	Numbers	Standard name
Logged job	The real training job used throughout: a Llama-2 70B LoRA fine-tune on 16 H100s, read from the GPUs' own power counters (section 1.2). Two further logged jobs, from a second lab, are used as checks in section 2.3	10.2 s step; 2.3–2.45 kW swing; one-second running means logged every 0.2 s, used at one a second	—
Injection	The logged job's power trace added, by arithmetic, to a recorded grid current	unscaled unless stated	signal injection
Unscaled	The trace is added as logged, not multiplied	scale factor 1	—
Campus data	The two grid records: LBNL (a 12 kV bus, 120 samples a second, 11.57 days) and Caltech (three meters from 252 kVA to 5.5 MVA plus an energy balance across one switchgear section; one sample a second, 7 days)	section 2.1	—
Island	The evader's training unit of at most 16 H100-equivalents (16 H100s in the case modelled here), wherever it connects. Rahman (2026) calls it a node; his nodes may also be, for example, 50 A100s	about 27 kW at the wall, an upper-end figure (section 0)	—
Premises	The customer whose meter the island sits behind: in the US, a home or small shop on single-phase service, or a larger business on three-phase	—	service point
Aggregation level	How far up the grid a meter sits, from one building to a campus supply line about the size of a feeder; measured here by the mean apparent power through the meter	252 kVA to 5,489 kVA for the pulse tests; step counts go down to 25 kW building mains	aggregation level; voltage level
Three-phase total	Sum of the three per-phase current magnitudes (per-phase kVA); tracks the three-phase apparent power a meter reports for a balanced load	—	—
Phase unbalance	One phase minus the mean of the other two, on current magnitudes. A balanced load cancels in it; a single-phase load does not	—	none exact; closest is "current unbalance". Negative-sequence current is the related phasor measure (Appendix A)
Band setting	The frequency range the spectral detector searches, with the segment length that sets its resolution. Two were run: **short-segment** (100 s segments, 0.08–0.5 Hz) and **long-segment** (500 s segments, 0.016–0.3 Hz). The logged job's 0.098 Hz falls inside the band at either segment length	—	search band; Welch segment length
Threshold	The worst hour without the job in the learning half, times a **margin**	margin 1 or 2	CFAR threshold with a multiplier

Term	Meaning	Numbers	Standard name
Time order	Learn on the earlier half of the record, score the later half. **Reverse** learns on the later half and is shown only as a check, because it has seen the future	—	walk-forward (out-of-time) validation
Energy balance	Power flowing into a section of switchgear minus power flowing out through every metered exit. Zero up to losses and meter error if every exit is metered and the meters are correctly scaled; an island inside the section shows as a leftover	one section measured: a steady +77 kW offset (losses), ±30 kW about it as found, ±11 kW (one standard deviation) after one meter was rescaled by a weight fitted on the same week	energy balance (Kirchhoff) across a bus
False alarm	An hour or an event the detector flags with no island present. The grid event behind it (a chiller start, a periodic load) is a **confounding event**	—	false positive

How to read a detection result. We take a week or more of recorded grid current, split it in two, let the detector learn the line's normal behaviour on the first half, and score the second half one hour at a time, twice: once as recorded, and once with the logged job's power trace added to it. A cell such as "100% (1/84)" means the detector flagged all 84 scored hours with the job added, and 1 of the same 84 hours without it. The second figure is the false-alarm count. A result is "at chance" when the two rates are about the same. A threshold such as "found in 80% of hours" is a convention, not a requirement: we treat any channel that flags the island more often than it raises false alarms as potential evidence. In principle evidence can add up over months and across channels, but only if false alarms are independent across hours and channels and their rate is known. Our data show neither, so the combined false-alarm rate is not established. More months also help only against confounding events that come and go; a premises load that switches on and stays on has to be resolved with other data.

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0. The question

- **The question this report tests.** Rahman writes: "If a frontier-scale cluster were split into nodes smaller than the regulator's monitoring threshold, it would be undetectable by thermal and electrical means, obligating enforcement to rely on network monitoring and in-person observations" (Rahman 2026, arXiv 2605.29359, Introduction, p. 1) [A-src]. His footnote 1 defines a node as "a single, self-contained unit of computing hardware" and sets the nodes he models below the hardware reporting threshold of Scher et al. (2025); his node is what this report calls an island. The sentence is conditional on the *monitoring* threshold; the footnote ties that to the *reporting* threshold. We test the second reading, and the electrical half only; we have no measurement of thermal detection.
- **The evader.** Rahman's run uses islands of at most 16 H100-equivalents. The islands talk over 100 Mbps links for up to 740 days. Run size, together with node hardware and training technique, sets the island count. At 100 Mbps the companion outline's runs of his model give two islands for a 10^{24} run, 34–168 for 10^{25}, about 1,100 for 10^{26} and 8,500–17,000 at frontier scale, the last two only with weight streaming (32,252 for 10^{26} without), all for a two-year run [S-sim: companion outline, section 2.4]. The paper's own Table 1 gives 2, 7, 34, 101, 625, 2,880 and 4,706 nodes for 10^{24}, DeepSeek-V3, 10^{25}, GPT-4, Llama 3.1-405B, GPT-5 and 10^{26}; the Llama row uses 50 A100s per node [A-chk: Rahman 2026, Table 1]. One island draws about 27 kW at the grid [A-arith: 16 × about 1.7 kW per H100, a datacenter planning figure equal to 22 kW IT at an overhead of about 1.2]. This is an upper-end figure: the logged job's measured GPU plus CPU draw is 10.9 kW [M], and a home or shop could run at an overhead from about 1.0 to above 1.5 (Uptime Institute 2024 gives 1.3 for efficient new builds, 1.56 on average) [A-chk], giving roughly 22 to 35 kW. A real island running this workload may draw less than the 25 kW used in the turn-on tests. The pulse results do not depend on this level: the job trace is mean-removed before it is added, so only its swing enters (section 2.1). The level matters for the turn-on and energy-balance results, so section 2.4 gives them at both ends of the range for this workload, about 11–27 kW [A-arith]. In the cost model each island's link is busy around the clock by construction [S-sim: an assumption, not an output].
- **The monitor.** A regulator, or a treaty party at home, installs monitors throughout the national electrical grid. Much of the hardware is already there: most feeders leaving a distribution substation have a digital protective relay that samples the phase currents, and usually the voltages, at about 2 kHz; about 40% of US feeder relays were still electromechanical in 2012 [A-chk: Newton-Evans 2014; SEL-751 and ABB REF615 data sheets]. What relays do not do today is record and report those samples continuously, and their metering is good only to about 1–5% on real power. The pulse results below used one-second means of each phase, the notch results 120 readings a second (Appendix A). The target is undeclared domestic compute on untracked hardware. The design does not resist a covert state. Registration is assumed at about

16-chip or 20 kW granularity, so chip accounting does not see the island. Of the remaining layers (network monitoring, the paper trail, in-person observation, thermal, energy accounting and grid-side monitoring), this report tests only the electrical ones.

- **Two placements.** (1) A home or small shop on single-phase service. (2) A three-phase commercial premises. Server power supplies take 200–240 V single-phase, so a 16-chip island does not need three-phase service.

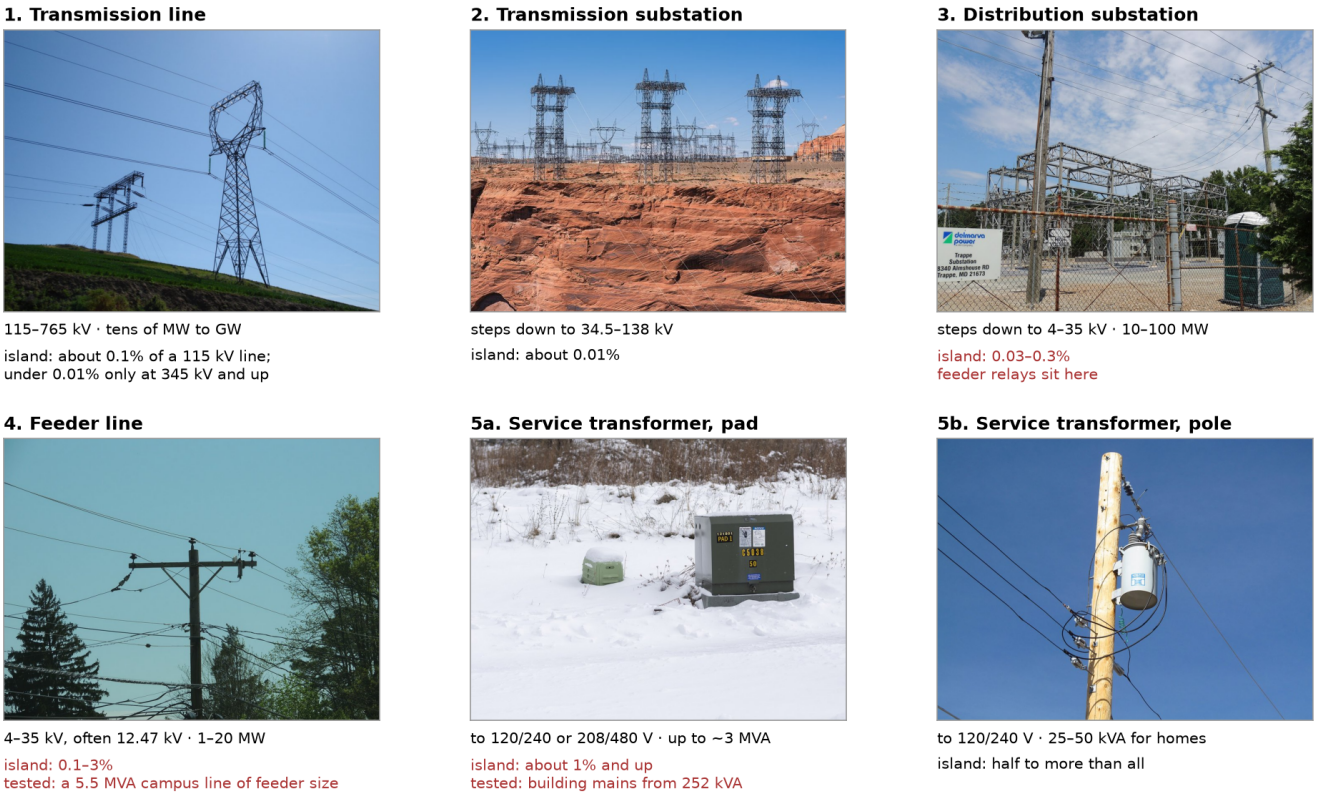


Figure 1. Where an island connects. The island is 27 kW at every level; its share of the load falls from about half or more of a home's service transformer to about 10^{-3} of a 115 kV line, and under 10^{-4} only at 345 kV and above. For a single-phase island, Part 2 finds the pulse at four meters from a 252 kVA building main to a 5.5 MVA campus supply line about the size of a feeder (marked "tested" in panels 4 and 5a; the fourth is the LBNL 12 kV bus); nothing below a building main was tested. Most relays on feeders are digital and sample the phase currents (Appendix A). Voltages and powers are agent recall [A-arith], except feeder sizes [A-chk: Short's table via electrical-engineering-portal.com; Iowa State EE455 slides]; Appendix A gives the same levels as a table. Photos: Wikimedia Commons (CC BY and CC BY-SA); credits under Sources.

Core result. For a 16-chip island on single-phase service, read on the phase unbalance, the logged job is found in every scored hour at each of four measured meters, from a 252 kVA building main through a 724 kVA bus and a 1.2 MVA section up to a 5.5 MVA campus supply line about the size of a distribution feeder (Figure 1, panels 3–4), with 1 to 4 false alarms in 84–121 hours at margin 1 and none at margin 2 except one at 5.5 MVA (section 2.3). Levels below a building main, and residential feeders, were not tested. Which of two pre-set band settings to report at each meter was chosen after scoring, and at 5.5 MVA the headroom is about 1×. A turn-on of 11–27 kW on one phase had no natural match of 10 kVA or more in 11.57 days on the LBNL bus. At the three Caltech meters a start or stop of one phase had no natural match of 11 kVA or more in a week at 252 kVA and 1.2 MVA, and none of 25 kVA or more at 5.5 MVA; a step marks when a load appeared, not that it trains (section 2.4). Spread evenly over three phases, the logged swing is weakly above chance at the building main (21% of hours against 2 false alarms in 84, margin 1 only) and at chance from 0.7 to 5.5 MVA; an island two to four times larger is found at 0.7–1.2 MVA. On the three-phase total a 25 kW turn-on had no natural match in a week at five of eleven building mains, none in a 2.4 kV energy balance only after one meter was rescaled on that same week (17 at face value), and about a hundred at a 5 MW line; at 11 kW, the low end of the island's draw, only two of eleven building mains had none and the corrected balance had 37 (section 2.4). A stationary false-alarm model misses by two to three orders of magnitude because the grid's background changes from hour to hour, so the rates above use thresholds learned from each meter's own earlier hours (section 2.6). Where three-phase service exists, the evader can defeat both single-phase results at no cost by balancing the load across phases. That takes deliberate

cabling: an improvised install keeps about 0.58 of the single-phase signal and is found up to 1.2 MVA by arithmetic on the stored results; a rack cabled per vendor guidance is found at a building main at best (section 3.1). The monitor's answer, a per-second premises meter or a balance across meters low in the network, is untested beyond the balance's background at one node for one week (Part 3). Every pulse result is for H100-era hardware. The headline rests on one Llama-2 70B LoRA fine-tune (batch 1 per GPU, CPU offload). Two further logged jobs from a second lab (Elsayed et al. 2026, one 8-H100 node each) were run through the same Caltech design. A Llama-8B run with a 13 s step is found in the same cells, needing 1.5–1.9 times the kW swing, as its short low phase predicts. A diffusion run with a 16 s step and the smallest swing of the three (2.2 kW) is found in every hour at margin 1 at all three Caltech meters on the long-segment setting, but at 5.5 MVA in only 18% of hours at margin 2; the short-segment setting misses it above the building main, because that setting searches for steps of 10 s or less [S-job]. At LBNL both are found in every hour on the phase unbalance, needing 0.26–0.42 of their own swing against the fine-tune's 0.20–0.24 [S-job]. A Stable Diffusion run from the first lab is found at LBNL through blocks its source attributes to evaluation (section 2.8).

Part 1. How an island could show up

Four signals. Part 2 gives what was measured for each.

1.1 Load against what the premises should draw

The island's energy is the simplest signal: 27 kW for months. Whether it stands out depends on what else sits behind the same meter, so this section follows the island from a home, where it is twenty times a household, to a shared feeder, where it is 0.4%.

- **What the signal is.** The island adds about 27 kW, an upper-end figure (section 0), all day, for months. The question is whether that is unusual for the meter that carries it.
- **At a home.** The island draws about 20 times an average US home's mean load (10,791 kWh in 2022, about 1.23 kW) [A-arith on A-chk: EIA]. Ordinary hourly smart-meter data would show it, given access and a baseline from before the island: a synthetic 12 kW load on hourly Spanish household data was found in every trial (section 2.2) [S-syn on M]. A battery cannot hide energy. 84% of US meters were advanced (AMI) meters at end-2024, up from 77% in 2023, and AMI need only record hourly [A-chk: EIA Electric Power Annual, Table 10.05]. That figure counts meters. It does not count regulator access to their data.
- **At a three-phase commercial premises.** 27 kW is ordinary as a level: an average US restaurant draws about 24 kW over the year [A-chk: EIA CBECS 2018, Table C22]. What is unusual is the shape: the island is flat around the clock, where most businesses drop outside opening hours. Comparison by business type or load shape: nothing measured.
- **On a shared line.** The island is about 0.4% of a typical feeder at peak [A-arith on A-src]. A mean shift of that size sits inside normal day-to-day variation (Part 2).

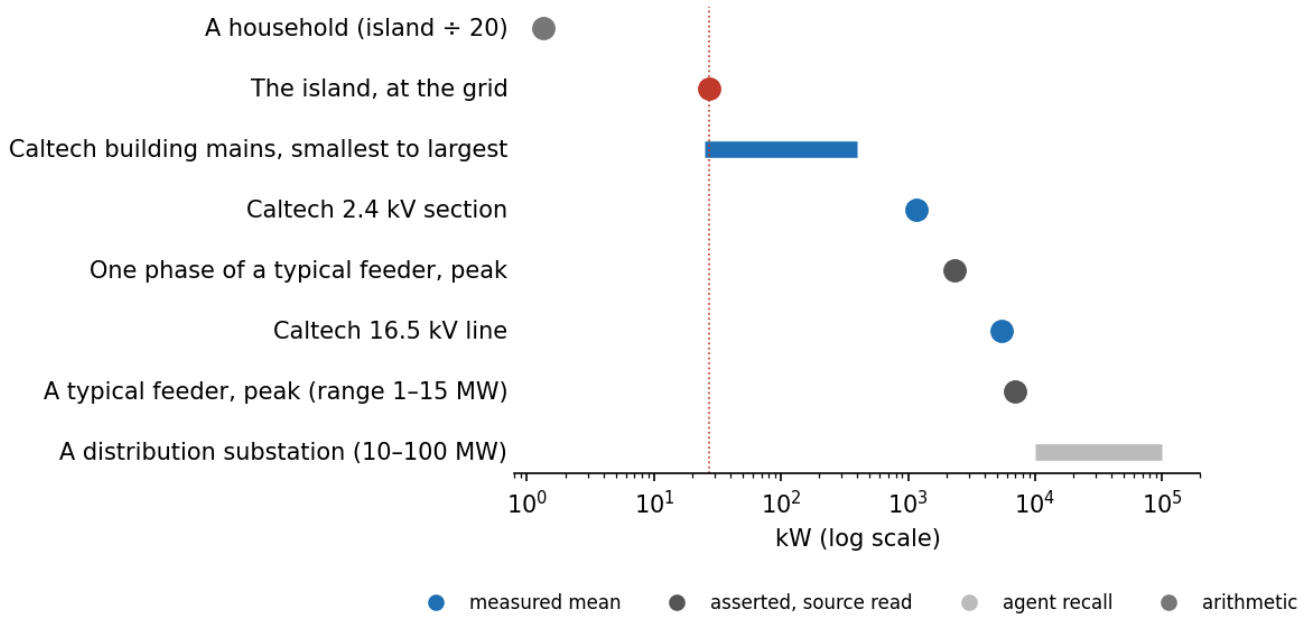


Figure 2. The island against the meters it could sit behind. Caltech values are measured weekly means [M] (building mains in kW; the 2.4 kV section, a net flow after on-site generation, and the 16.5 kV line in kVA); feeder figures are [A-chk]; the substation range is agent recall [no source found]; the household is the island divided by 20 [A-arith], close to EIA's 2022 average of about 1.2 kW.

1.2 Pulsing during training

Training is not a steady load. In synchronous data-parallel training each step alternates a compute-heavy phase with a lower-power one (communication, and in the logged job also CPU offload traffic), and GPU power follows, so the island broadcasts a periodic signal at its step frequency. Most of Part 2 rests on this signal, and this section describes the main job on record and two checks.

- **What the signal is.** Each training step has a compute-heavy phase and a lower-power phase. GPU power rises and falls with them; overlapping communication with compute shrinks the low phase. From the grid side this is a periodic load signal at the step frequency and its harmonics, the kind that forced-oscillation detectors (Follum & Pierre) are built to find.
- **The main logged job.** A Llama-2 70B LoRA fine-tune (MLPerf, DeepSpeed ZeRO-3 with CPU offload) on 16 H100s (4 nodes), batch size 1 per GPU, five replicate runs (Vercellino et al., NLR data set, DOI 10.7799/3025227). Its step is a near-square pulse. The step time is 10.1–10.3 s, with 5.6–5.8 s high and 4.4–4.6 s low. The high level is 10.97–11.00 kW. The power swing is 2.32–2.45 kW [M]. All 16 GPUs move together: the total swing is 0.96–0.97 of the sum of the GPUs' own swings [M].
- **What the log is.** Software readings of GPU power, logged every 0.2 s, each a mean over the previous second. The paper calls them instantaneous. The tool's commit history and NVIDIA's documentation say otherwise [A-src: WattAMeter commit `984f6f9`; five-sample ramps in the data]. The authors have not been asked. The GPU readings include the GPU's own memory, and CPU package power is logged separately. Host memory, NVSwitches, network cards, storage, fans and power-supply losses are absent.
- **Size at the wall.** The swing is 21–22% of GPU power. It is about 9% of the island's 27 kW at the wall [A-arith].

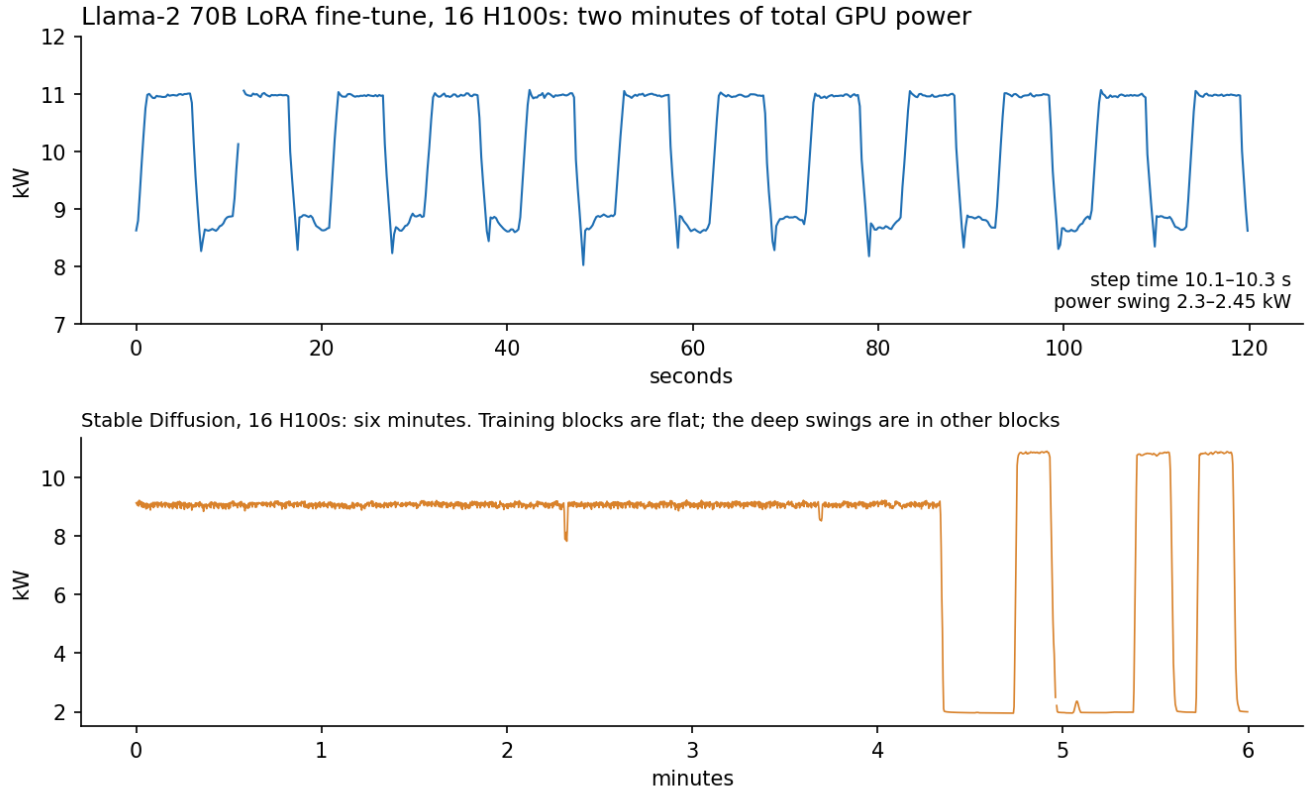


Figure 3. The two logged workloads [M]. Top: two minutes of the fine-tune; the step is a near-square pulse. Bottom: six minutes of Stable Diffusion; long blocks are flat at one-second resolution, and the deep swings come in other blocks. The source paper attributes the flat blocks to training and the swings to evaluation; the logs carry no phase labels [A-src]. In one of five runs most blocks swing. Both are software GPU readings, each a mean over the previous second, not wall power.

- **Other cadences.** Reported step cadences run from about once a second to once every tens of seconds (about 0.02–1 Hz), with spectral energy to about 3 Hz from the pulse shape [A-chk: Choukse et al. 2025, arXiv 2508.14318, one at-scale DGX-H100 job]. A 0.05 Hz figure (Ko et al. 2025) reuses Choukse's trace; grid modelling studies assume 0.5–1.5 Hz [A-rep]. Apart from the logged jobs here, the sources describe at-scale jobs or give no size. We found no pre-training cadence measured on a 16-chip island, and have not checked single-node traces such as Latif et al.
- **Two checks from a second lab.** Elsayed et al. (2026) publish 15-minute per-GPU power logs of training sessions on single 8-GPU cloud nodes. We use two H100 sessions [M]. A Llama-8B run (ZeRO-3, no CPU offload) steps every 13.4 s with a 3.7 kW swing and is low for about 14% of each step. A diffusion run (107M parameters, 32-pixel images) steps every 16.1 s with a 2.2 kW swing and is high for about a third of each step; it pulses steadily through the whole session. The logs are labelled 20 ms readings but change like one-second means, as the fine-tune's do, so they are good for the step frequency and not for the edges. Host CPU load is 4–6% and does not follow the step.
- **What has been measured before.** We are not aware of a prior measurement in which a training-step cadence was recovered from a utility-side current with the job buried in other load. The nearest work is at the chip or its power cable, synthetic (Kimpson 2026, a detector bank run on modelled scenarios) [A-src], or at facility scale, where the data center is most of the load and the recorded oscillations are converter or UPS modes [A-rep: prior-art search of 2026-10-09; NERC Large Loads Task Force 2025; PNNL-39191]. Koopmanschap & Barten (2026) state "There is no externally distinguishing power signature at this scale" for 100 H100-equivalents, an argument from load level alone [A-src: arXiv 2607.22619, Appendix B]; the sentence from Rahman (2026) quoted in section 0 is the other such statement we found. The periodic-load picture itself is not new (Choukse et al. 2025).
- **Distributed training keeps the step.** Inside each island, DiLoCo-style training is ordinary synchronous training [A-src: DiLoCo, arXiv 2311.08105; DiLoCo scaling laws, arXiv 2503.09799; INTELLECT-1, arXiv 2412.01152; Covenant-72B, arXiv 2603.08163]. So each island keeps whatever step swing its own setup produces, which may be smaller if in-island communication is fast or overlapped. Blocking outer exchanges add a slower on/off swing. We found no published power measurement of any DiLoCo-style run.

1.3 On/off steps

Energy is a level; a turn-on is a change. Changes are easier to see than levels, because a meter's fixed offset cancels in a before-and-after difference and its gain error only rescales the step by a few percent. The question for this signal is how often the grid produces a 25 kW step of its own at the place the island connects.

- **What the signal is.** The island has a turn-on and a turn-off. A run that pauses to exchange results has a restart step at each exchange. A crash or stall has a dip and a restart.
- **Why it can work.** A step is a change. Fixed offsets and steady losses cancel in a before-and-after difference. A fixed gain error only rescales the step by the same few percent (3% of 27 kW is under 1 kW), which does not hide it. So for a step, a meter's noise and resolution matter more than its accuracy class. Those of installed relay metering for a 27 kW change are not published and have not been measured.
- **Why it can fail.** Buildings switch large loads on and off all day. The question is how often a natural step of 25 kW or more occurs where the island connects.
- **How often an island stops.** One 16-chip island is interrupted about 0.4 times in 54 days, roughly once in 130 days, if Llama 3's rate scales linearly per GPU [A-arith on A-chk: Llama 3 reports 466 interruptions, 419 unexpected, in a 54-day snapshot on up to 16K GPUs (Grattafiori et al. 2024, arXiv 2407.21783)]. Network and software faults need not scale this way, and an island on less managed hardware may fail more often, so this is closer to a lower bound. Crash steps are rare. Planned exchange steps recur at every exchange if the run pauses for them (section 3.6).

Figure 3 shows what a training run looks like, as its GPUs report it. Figure 4 shows what the grid does on its own, as a utility meter records it: one motor start on a real bus. They differ in three ways. The training pulse repeats every 10 s for as long as the run lasts; the motor start happens once. The island is small against the bus: its turn-on is about 3.7 A on one phase and its pulse about 0.34 A, where the motor start moves every phase by several amps. And a motor draws on all three phases together, so it cancels in the phase unbalance, while an island wired to one phase does not (section 3.1 covers balancing it across three). The repetition is what an hour's spectrum picks up, and the phase difference is why the island is found on the phase unbalance and not on the total (section 2.3).

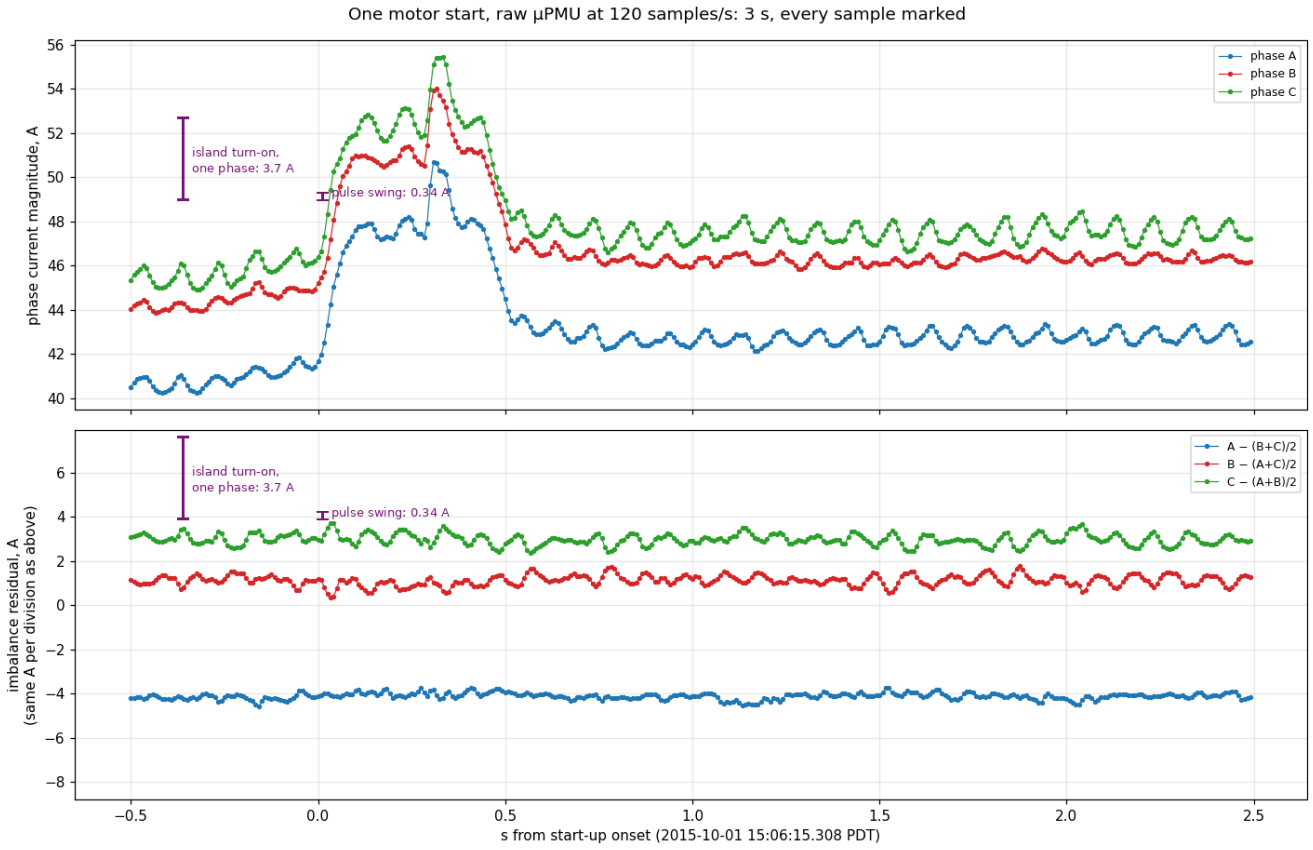


Figure 4. One motor start on the LBNL 12 kV bus, every sample at 120 a second [M]. Top: the three phase currents all rise together, by about 6 A each in the first quarter-second and about 9 A at the peak, then settle about 1.5 A above where they started (about 30–40 kVA in all). Bottom: the phase unbalance (each phase minus the mean of the other two) barely moves. A balanced three-phase load cancels in the phase unbalance; a single-phase island does not. The purple bars give the island for scale: at 7.2 kV phase-to-neutral, its 27 kW turn-on on one phase is about 3.7 A and the logged job's 2.45 kW pulse swing is about 0.34 A [A-arith; bars added to the results figure by `src/annotate_fig03.py`*]. The turn-on would be a plain step in the lower panel; the pulse is about the size of the 10 Hz ripple on that panel (about ±0.4 A), which is why it is usually found in an hour's spectrum and not by eye (Figure 5 shows a quiet hour, where the eye works too). This cancellation is why the phase unbalance is quieter than the total on both campuses.*

Figure 5 combines the two: the logged fine-tune's pulse from Figure 3, converted to current, added to one phase of ten minutes of the bus in Figure 4.



Figure 5. Ten minutes on the same bus, every sample at 120 a second, with the logged fine-tune's pulse added to phase 1 from 150 to 450 s (shaded) [S-job; kW / 7.2 kV = A, in phase]. The window (2015-10-10, 05:54 PDT) was drawn at random and fell in a quiet hour: the background's one-second phase unbalance varies by 0.07 A (standard deviation) against the job's 0.34 A swing. Top: the three phase currents; the job cannot be seen. Second: the phase unbalance, every sample; the job shows as bursts in the shaded span. Third: one-second means; the square wave is plain. Bottom: the job alone. In four more random ten-minute windows, two of them weekday daytime, the background varies by 0.14–0.28 A and the job is hard to see by eye without the background-only trace beside it; the spectral detector finds it in every hour either way (section 2.3).

1.4 Harmonics (multiples of 60 Hz)

A repeating on/off pulse (such as the top panel of Figure 3) is, mathematically, a sine wave at its own frequency plus weaker sine waves at whole multiples of it, called harmonics; only a pulse that is on exactly half the time lacks the even ones. The same idea applies at 60 Hz, though the two sets of harmonics are unrelated: an uncorrected rectifier draws current from the line in pulses, so its current carries harmonics of 60 Hz. Server supplies correct their input, which keeps those harmonics to a few percent. This section asks whether a 16-chip island adds enough at those frequencies to stand out from a building already full of motor drives and converters. The answer depends on which harmonic: the low ones are lost, the 11th and 13th may not be at a small building main (Figure 10).

- **What the signal is.** Three-phase (six-pulse) rectifiers and drives put current at 300, 420, 660, 780 Hz and higher (5th, 7th, 11th, 13th). Single-phase rectifiers also put current at 180 Hz, 540 Hz and the other odd multiples.
- **Why it may not work.** Modern server supplies correct their input; many UPS units do too, but older six-pulse-rectifier units do not. A corrected input has a total current distortion of about 1.5–3% at medium to full load, rising to 7–14% at light load [A-chk: TI TIDA-00707]. Its switching frequency (100–140 kHz in one TI reference design) puts its distinctive emissions in the 2–150 kHz band [A-src; no measurement of a server supply found]. The meters in hand sample at 2.5 kHz, so they resolve up to about 1.25 kHz, one second in every ten.
- **Figure 10** (section 2.5) shows the measured background against what an island would add.

Part 2. Results

2.1 Data and method

Part 2 reports what was measured for each of the four signals. This section names the two grid records, how the logged job was added to them, and how the detector's thresholds were set, so that the rates in sections 2.2 to 2.6 can be read without the results files.

- **LBNL.** Open µPMU data set (Stewart et al. 2016), 12 kV bus `grizzly_bus1_2`, 2015-10-01 to 10-12 (11.57 days), 120 samples a second. Analysed as 1-second means (the other five runs in section 2.3 used a low-passed 12-a-second series). Mean load 724 kVA on the 1-second pipeline [M].
- **Caltech.** SoCal 28-bus data set (arXiv 2504.06588), eGauge meters, 2025-03-10 to 03-17 (7 days), one sample a second. Three meters: a 252 kVA building main (480 V), a 1,160 kVA 2.4 kV section and the 16.5 kV campus supply line at 5,489 kVA, plus an energy balance computed across the 2.4 kV switchgear from several meters [M; 252 kVA recomputed from re-pulled raw data, the others read from stored results]. The 2.4 kV figure is a net flow: the section's buildings draw about 1.66 MW, about 0.7 MW of it from on-site generation [M]. The 16.5 kV meter is the supply into the campus substation; about 3.9 MW of its load is one lump with no meter under it.
- **Injection.** The logged job's power trace, cut to its busy stretch, mean-removed, resampled and looped to fill each hour, is added to the grid record at its logged size, kW taken as kVA [S-job]. "Single-phase island" means the job is added to one phase. The monitor then reads the phase unbalance: one phase minus the mean of the other two. "Three-phase island" means the job is split evenly. The monitor then reads the three-phase total.
- **Thresholds.** The threshold is the worst no-signal hour of the learning half, times a margin of 1 or 2. Time order (out-of-time validation) learns on the earlier half and scores the later. The reverse learns on the later half; it has look-ahead bias and is shown only as a check. Windows are one hour. Spectral detector, run with two band settings: short-segment (100 s segments, 0.08–0.5 Hz) and long-segment (500 s segments, 0.016–0.3 Hz); see Terms. Both were fixed before the job was injected (copied from LBNL); which one to report at each meter was chosen after scoring.
- **Source of the detection rates in 2.3 and Appendix D.** `task3_caltech_real_job.json` and `task2_slow_pulse.jso` re-tabulated by `src/tabulate_rates.py` (`src/rates.md`). No new run. The 2.2 rates come from `20260823_s3_detection_roc.md`; the 2.4 counts from `caltech_checks.json`, `coincidence_caltech.json` and the held-out measurement files.

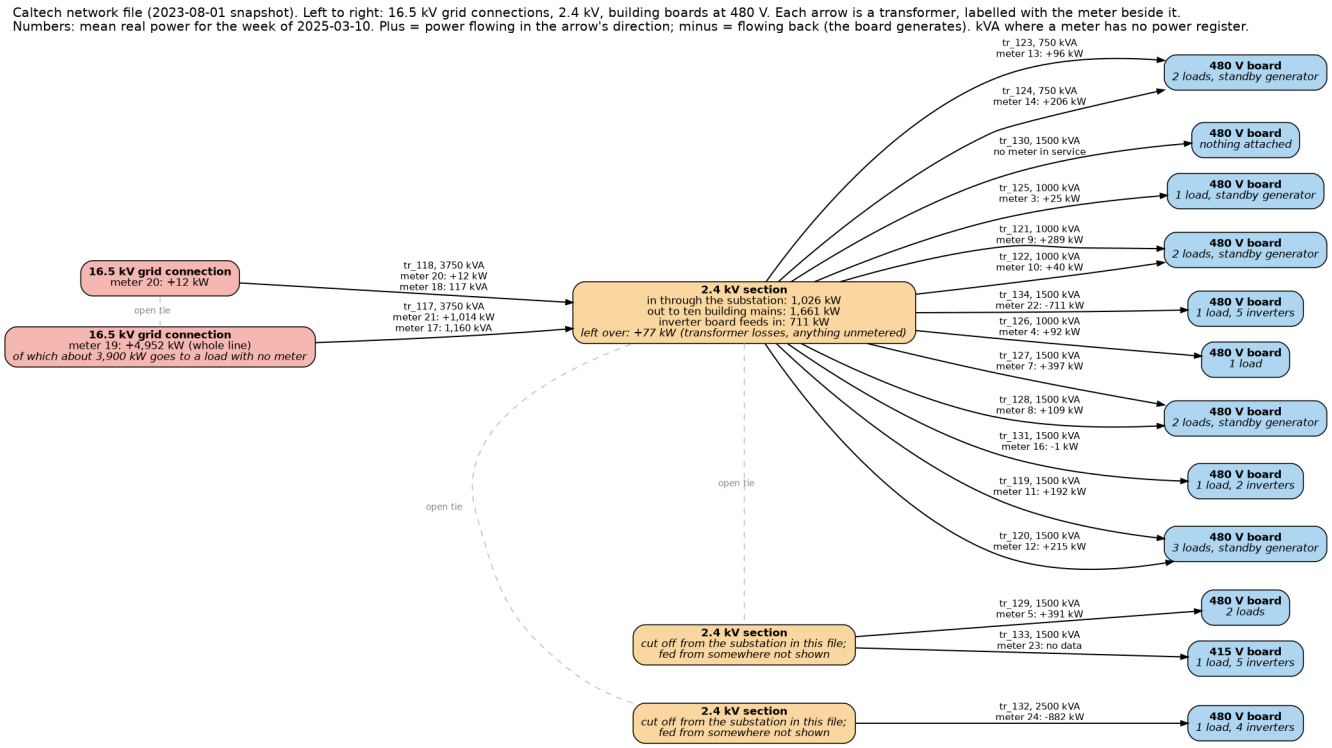


Figure 6. The Caltech network file (a 2023 snapshot) drawn as a tree: 16.5 kV grid connections, the 2.4 kV section, and the 480 V building boards. Each arrow is a transformer, labelled with the meter beside it where there is one; numbers are mean real power for the week, or kVA where a meter has no power register [M]. Two further 2.4 kV sections are cut off from the substation in this file. The energy balance in section 2.4 is taken across the main 2.4 kV section.

2.2 Load against what the premises should draw

Results for the island's energy come from synthetic constant loads added to measured premises and substation records, not from the logged job. They show where in the network a steady 12 to 100 kW is visible by itself, and they stop where the shared feeder begins.

- **At a single premises.** A constant 12 kW load is found in every trial at a 1% false-positive rate (300 trials, 60 Spanish supply points, mostly households, of 0.28 kW mean and 0.19 kW median, GoiEner data, each scored against its own history) [S-syn on M].
- **At a 1–5 MW zone substation.** Energy mean shift on 15-minute data is at chance. The area under the curve is 0.50 for 12 kW and 0.52 for 100 kW. At a 1% false-positive rate, 100 kW is found in 1.2% of trials (480 trials, Ausgrid data, about 3.2 MW mean, 30-day horizon) [S-syn on M]. Trials overlap, so they are correlated.
- **Between the two (synthetic feeders).** At 200 premises, 12 kW gives an area under the curve of 0.84 and is found in 20% of trials at a 1% false-positive rate. At 1,000 premises the area is 0.58 [S-syn on S-syn]. The middle tiers are synthetic feeders built from measured premises.
- **Against business type.** We have no measurement of how an island's load compares with what its declared business type should draw.

2.3 Pulsing during training

The main result. The logged job's trace was added, unscaled, to both grid records, once on one phase and once split across three, and the spectral detector scored every hour in both directions of time. Two tables give the rates; the bullets say what survives in time order.

Single-phase island (phase unbalance). Logged fine-tune, first run, unscaled [S-job]. One row per meter, at the band setting that held at both margins, chosen after scoring; both settings in both directions are in Appendix D. Cells: found in % of hours (false alarms / hours scored).

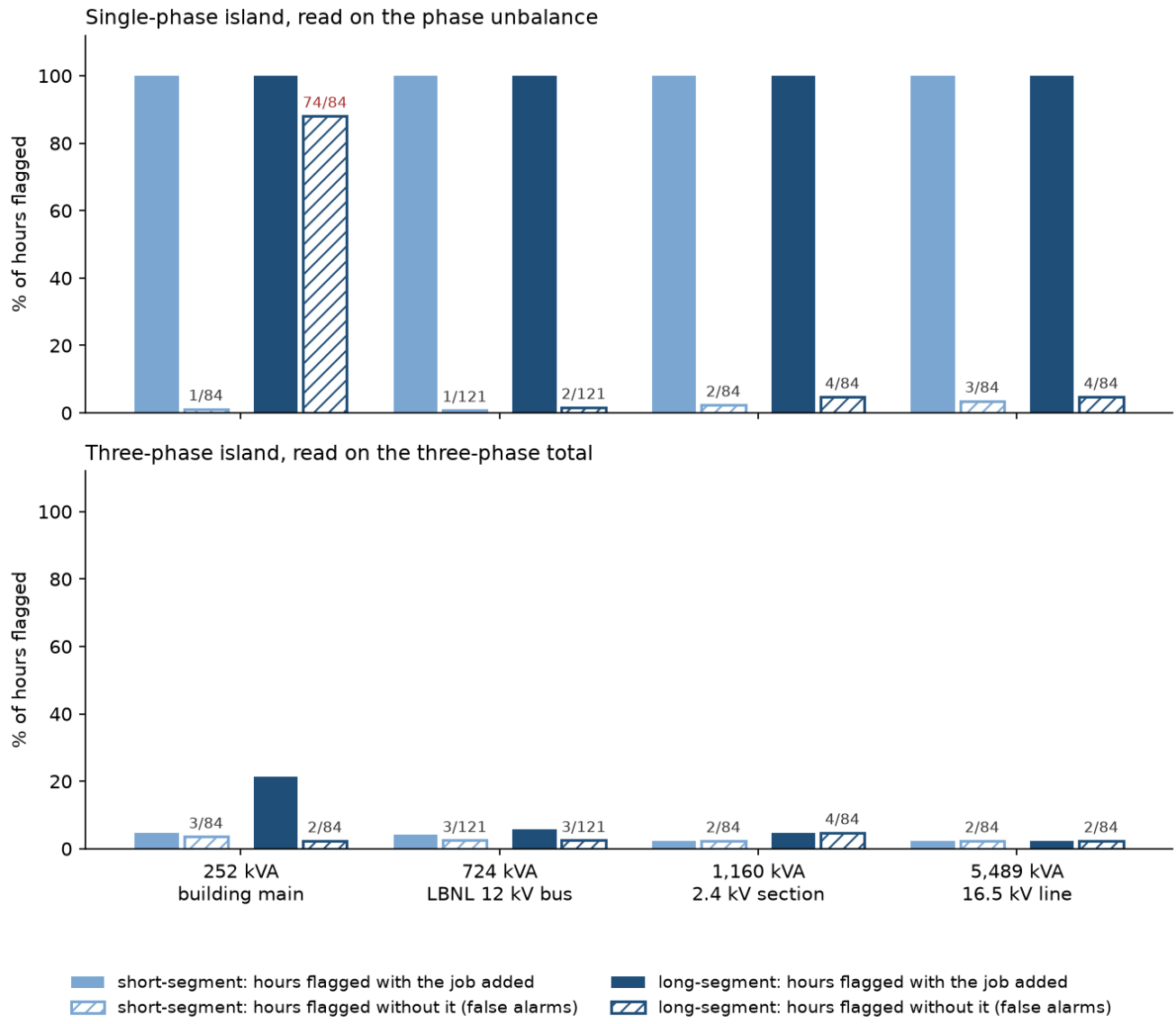


Figure 7. The logged job, unscaled, by aggregation level [S-job]. Top: single-phase island, read on the phase unbalance. Bottom: three-phase island, read on the three-phase total. Filled bars: share of hours flagged with the job added. Hollow bars beside them: share flagged without it (false alarms), same hours. Time order, margin 1, both band settings. Where the hollow bar reaches the filled one the result is at chance (the long-segment setting at the 252 kVA building main, top).

Aggregation level	Setting	Time order, margin 1	Time order, margin 2
Caltech building main, 252 kVA	short-segment	100% (1/84)	100% (0/84)
LBNL 12 kV bus, 724 kVA	short-segment	100% (1/121)	100% (0/121)
Caltech 2.4 kV section, 1,160 kVA	short-segment	100% (2/84)	100% (0/84)
Caltech 16.5 kV line, 5,489 kVA	long-segment	100% (4/84)	100% (1/84)

- At every meter the job is found in every hour, with 1 to 4 false alarms in 84–121 hours at margin 1 and none at margin 2 except one at 5.5 MVA.
- The other setting is not always clean. At the building main the long-segment setting is at chance (74 false alarms in 84 hours at both margins, Appendix D); the likely cause is spectral lines that appeared or moved in the test half, not checked line by line. At 5.5 MVA the short-segment setting finds the job in every hour at margin 1 in time order but in 45–95% at margin 2 (the range spans the three phase residuals), and in reverse in only 0–6% even at margin 1. So the setting differs by meter, and which one to report was chosen after scoring (limit 6).

- Headroom shrinks with level. In time order at margin 2, a swing of 0.20–0.29 times the logged swing is enough to find it in most hours at LBNL (0.20–0.33 across all six logged runs) [S-job]. At Caltech the fraction needed is about 0.3–0.7 at the two smaller meters and 0.5–1.1 at 5.5 MVA, so the headroom there is about 1×. "Most hours" is the 80% convention, given here only as a measure of headroom.
- The four other fine-tune runs and the Stable Diffusion run behave alike at LBNL: 100% of hours, 0–2 false alarms in 121 [S-job; low-passed 12-a-second series]. These are two workloads, not six. Stable Diffusion is presumably found through its swinging blocks, which the source paper attributes to evaluation; its long blocks are flat at one-second resolution [M]. The detector ran on the whole trace, not block by block.
- **Two jobs from a second lab, at Caltech.** The two Elsayed et al. sessions (section 1.2) were added, unscaled, through the same design at the three Caltech meters [S-job]. The false-alarm counts are identical to the fine-tune's in every cell, as they must be, which also reproduces this section's false-alarm figures from freshly downloaded copies of the Caltech data.
  - The Llama-8B run is found in 100% of hours in every cell where the fine-tune is, at both margins. It needs 1.5–1.9 times the fine-tune's swing in kW to be found in 80% of hours, which its short low phase predicts: the detector keys on the step frequency, and a short dip puts less of the swing there. Measured against its own swing the headroom is about the same as the fine-tune's.
  - The diffusion run, the smallest swing of the three (2.24 kW), is found in 100% of hours at margin 1 in time order on the long-segment setting at 1.2 MVA and 5.5 MVA, and on the short-segment setting at 252 kVA. At 5.5 MVA, margin 2, it is found in 18% of hours in time order (58% in reverse); it needs 1.55–1.78 kW at margin 1 there, so its headroom is about 1.3 times. On the short-segment setting it is found in 81% of hours at 1.2 MVA at margin 1, none at margin 2, and in 2% at 5.5 MVA. That setting searches for steps of 2.5–10 s; this step is 16 s and is seen there only through its weaker harmonics.
  - So which setting works depends on the job's step time as well as on the meter, and a monitor does not know the step time in advance. Running both settings is the obvious remedy and roughly adds their false alarms (not computed).
- **The same two jobs at LBNL** [S-job; added 2026-10-15]. Both are found in 100% of hours on the phase unbalance, on both settings, in both directions and at both margins; the false alarms are the fine-tune's, since they are scored on the same no-signal hours. Neither is found on the three-phase total at its logged size (at most 5% of hours). On the long-segment setting at margin 2, the swing needed to find the job in 80% of hours is 0.26–0.37 of its own swing for Llama-8B and 0.29–0.42 for diffusion, against 0.20–0.24 for the fine-tune. The weak cell is again the diffusion run on the short-segment setting, at 0.59–0.96 of its swing. The fine-tune's LBNL cells were re-run first from freshly downloaded data and reproduce the numbers above exactly.

Three-phase island (three-phase total). Same job and design [S-job]. In time order the long-segment setting needs a smaller swing on the total at every meter; at the logged size both settings are at chance above 252 kVA, and in reverse at the building main the short-segment setting does better (Appendix D).

Aggregation level	Setting	Time order, margin 1	Time order, margin 2
Caltech building main, 252 kVA	long-segment	21% (2/84)	0% (0/84)
LBNL 12 kV bus, 724 kVA	long-segment	6% (3/121)	0% (0/121)
Caltech 2.4 kV section, 1,160 kVA	long-segment	5% (4/84)	0% (0/84)
Caltech 16.5 kV line, 5,489 kVA	long-segment	2% (2/84)	0% (0/84)

- At the 252 kVA building main, long-segment setting, margin 1, time order: the job is found in 21% of hours (about 18 of 84) against 2 false alarms. That is above chance at a low rate. The reverse direction gives 10% against 5.
- At 0.7–5.5 MVA the job is at chance on the total: found in 0–6% of hours against 0–4 false alarms. An island about 2 to 3 times larger is found in most hours at 0.72 MVA, 3 to 4 times at 1.2 MVA, and about 11 times at 5.5 MVA [S-job; time order, margin 2, no false alarms; reverse direction worse at Caltech].
- At margin 2 the long-segment rate is zero at every meter. The only nonzero margin-2 cells in Appendix D equal their false-alarm counts.
- The training pulse's own harmonics (0.2, 0.3 Hz and above) do not close this gap above 0.7 MVA. A square pulse keeps about 80% of its variance in the fundamental. Summing ten harmonics gains at most about 2 times in amplitude over the best single bin, most of it from the fundamental's second bin [M, LBNL, 600 s segments], against a gap of 1.6 times (252 kVA) to 11 times (5.5 MVA) the logged swing [S-job; long-segment, time order, margin 2; reverse worse at Caltech]. At the building main an ideal comb might close the gap. No harmonic comb was built.
- We did not test the job at a premises where it would be most of the load: Caltech building mains of 5–99 kVA have measured backgrounds but were not used in an injection run, and no utility revenue meter was tested. If the island is most of a premises' load and the rest of that load is steady, its swing should stand out at a meter that samples once a

second or faster [A-arith, not computed]. That is not guaranteed: one 59 kVA Caltech building has a 20 kW motor that starts about 50 times a day. Most US premises meters record every 15 or 60 minutes at best [A-rep]. So the first open question there is meter speed.

- A small host building is unmeasured; section 3.11 gives an extrapolation.

What a harmonic comb might show, if investigated [S-syn; added 2026-10-15]. The logs cannot show the pulse above about 0.5 Hz, because each reading is a one-second mean. To see what a µPMU's 120-a-second current magnitudes could show, we rebuilt the fine-tune's pulse as a two-level wave with instant edges at its own 578 logged edge times, which keeps its real step-to-step jitter, and added it to the LBNL phase-1 unbalance for five minutes. Averaged like the logger, the rebuilt pulse reproduces the logged spectrum up to about 1 Hz. Spectra use 60 s Hann windows (about six steps) at 50% overlap.

- In a quiet hour the job's harmonics stand clear of the background at most multiples of the step frequency up to about 1–2 Hz. In a busy weekday hour the fundamental and a few harmonics do (Figure 8).
- Above about 2 Hz the step-to-step jitter smears the lines, and longer windows stop helping.
- Edge shape matters little for what is shown. Rises on a millisecond scale weaken only harmonics above about 5 Hz. Only ramps lasting about a second (section 3.3) would weaken the comb above about 0.5 Hz.
- This is a simulation on real background, not a measurement, and no harmonic-summing detector was run. A wall-side trace of a training job would test it (Part 4).

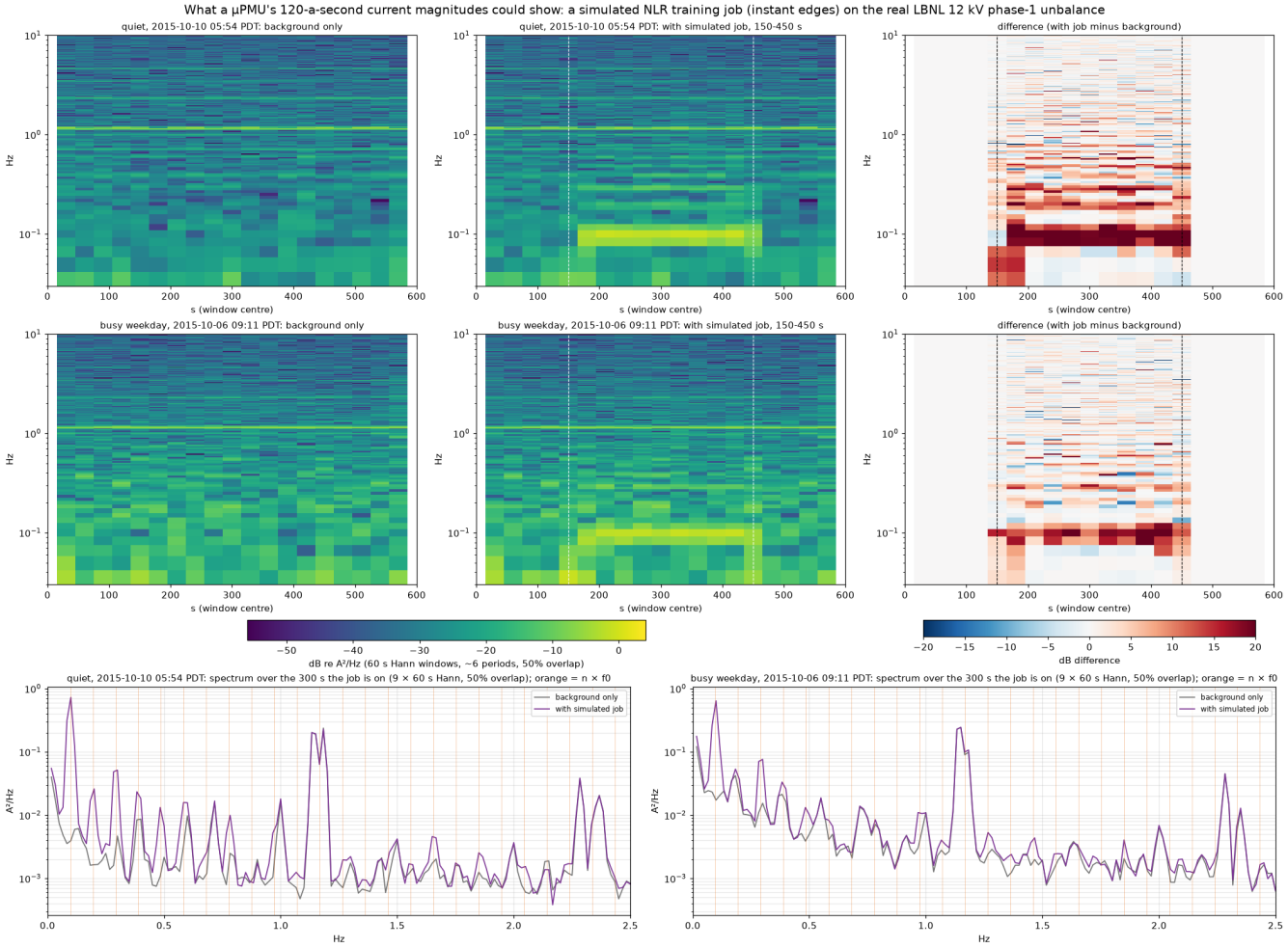


Figure 8. Simulation on real background [S-syn]: the logged fine-tune's step rebuilt with instant edges and added to the LBNL phase-1 unbalance from 150 to 450 s, as a µPMU's 120-a-second current magnitudes would show it. Rows: a quiet hour (05:54, the window of Figure 5) and a busy weekday hour (09:11). Columns: background only, with the job, and the difference in dB. Bottom: the spectrum over the 300 s the job is on, with and without it; orange lines mark multiples of the step frequency. The background's own lines at about 1.17 and 2.3 Hz are not the job.

Edge detector on the logged job. 0% of hours, 0 of 121 false alarms on the total [S-job]. This does not test the method. One-second means smear the edges it looks for.

2.4 On/off steps

For on/off steps the main measurement is a count: how often the grid itself produces a 25 kW turn-on, a pause or a short notch with no island present, since each is a false alarm a step detector would raise. One synthetic injection at the end tests restart-edge recovery.

What the island's step looks like. When training starts, a server with power-factor-corrected supplies raises its real power with almost no reactive part above about a quarter load, and the higher level holds; switching the supplies on adds a brief inrush, limited to a few cycles [A-chk: TI TIDA-00707]. A motor started directly across the line draws about 5–8 times its running current for a fraction of a second (NEMA Design B up to about 6.4 times) and takes reactive power for as long as it runs (Figure 4; Appendix A). A motor behind a variable-speed drive does not: it presents a power factor near 1 and ramps up without inrush, which looks like the island's step [A-chk: EC&M; Hoevenaars 1999]. Storage water heaters and stoves are pure steps of a few kW, but whole-house tankless water heaters are resistive steps of 27–36 kW [A-chk: Eemax EEM24036]. So the question is how often the grid produces a resistive step of 25 kW or more where the island connects. Across levels, a meter that carries more load sees more such jumps, though building to building the count varies more than load explains (Figure 9).

- **On the phase unbalance, at LBNL.** No natural up-step of 10 kVA or more in 11.57 days; 13 of 5 kVA or more [M, sizes fixed in advance]. A single-phase island's turn-on, anywhere from 11 to 27 kW, would therefore have had no false alarm in eleven and a half days on this 724 kVA bus. This is the cleanest step result we have, and it needs no fitted correction.
- **On the phase unbalance, at Caltech.** The same count, for up-steps (a start) and down-steps (a stop), on each phase's unbalance at the three meters of the pulse test, merged over the three phases since a monitor does not know the island's phase [M; one week, 5-, 15- and 60-minute windows].

Meter	Starts / stops of 11 kVA or more in the week	25 kVA or more	Largest natural step	Three-phase total, 11 kW or more
252 kVA building main	0 / 0	0 / 0	2.4 kVA	4 / 2
1.2 MVA, 2.4 kV side	0 / 0	0 / 0	8.2 kVA	136 / 138
5.5 MVA campus supply line	8 / 13	0 / 0	15.5 kVA	157 / 167

  Counts are for the 15-minute window; the 5- and 60-minute windows give the same picture. The largest step is over the 5- and 15-minute windows; at 5.5 MVA the 60-minute window reaches 20.8 kVA up and 24.6 kVA down, still under 25. The last column is the same count on the three-phase total in kVA (starts / stops), close to but not the same series as the real-power count in the next bullet.

  - The phases are measured separately (their means differ and their minute-to-minute changes correlate at 0.84–0.97, not 1). The campus's large swings move all three phases together, so the phase unbalance cancels them.
  - So a single-phase island's start and stop each stand out at all three meters. At 252 kVA and 1.2 MVA they stand out across the whole 11–27 kW range. At 5.5 MVA they stand out at 25 kW; an 11 kW start there sits below the largest natural step and has one or two look-alikes a day.
  - **What a step does not show.** A step says that a single-phase load of that size appeared, not that it trains. A Level 2 electric-vehicle charger is 7–19 kW on one phase and would be a common look-alike on a residential feeder; none appeared behind these meters in the week. A step is a cue in time for the pulse search, not evidence of training on its own.
  - It needs the same single-phase install as the pulse, so balancing across phases removes both together (section 3.1). With the improvised install's 0.58, an 11 kW start is about 6.4 kVA, which has about 40 look-alikes a week at 1.2 MVA and about 140–160 at 5.5 MVA. A start ramped over longer than an hour is not a step by this rule (section 3.9).
  - None in a week bounds the rate below about 3 a week at 95%. The three meters are on one campus, probably on one supply path, and the week is the one used for the pulse test. No step was injected.

- **Steps with no reactive part, at LBNL.** Natural steps of 5 kW or more with almost no reactive power (reactive change at most a tenth of the real change): about 2 to 5 a day per phase in each direction, against 170–240 a day of all kinds [M; three phases, 11.57 days]. The same real-only test passes 72% of planted island edges [S-syn].

- **Short notches, at LBNL.** Notches of 8–500 ms and 15 kVA or more: hundreds to thousands a day on the three-phase total, none on the phase unbalance [M; 3.1 days, 120 samples a second].
- **The turn-on, on the three-phase total, at Caltech.** False-alarm count: mean of the next 15 minutes minus mean of the previous 15 minutes, on 1-minute real power, over one week. No island was present, so every jump of 25 kW or more is a false alarm [M; thresholds read from the same week]. The island may draw as little as 11 kW (section 0), so the same count was repeated at 11 kW on freshly downloaded copies of the week, after checking that it reproduces the 25 kW column exactly [M].

Where	Mean load	Natural jumps of 25 kW or more in the week	11 kW or more
Energy balance across the 2.4 kV section, one meter rescaled (see below)	about 1,700 kW through the section (3,400 kW summed over meters in and out)	0 (largest 23 kW; 2 at a 5-minute window)	37 (78 at a 5-minute window)
Same balance, every meter at face value	same	17	82
Five building mains	25–215 kW	0	0, 0, 3, 6, 19
Three building mains	92–289 kW	1 each	1, 3, 66
Three building mains	192–397 kW	2–9	12–51
Substation meter	1,014 kW	16	125
16.5 kV line meter	4,952 kW	107	153

  - An energy balance should close to a steady offset (transformer losses and anything unmetered) plus meter error (Terms). This one sits at a steady +77 kW and varies around it by ±11 kW (one standard deviation), 0.3% of the 3.4 MW summed over its meters or about 0.6% of the roughly 1.7 MW passing through the section. That holds only after one meter, on the solar-and-plant switchboard, is counted at 0.58 of its reading, a weight fitted together with the offset on the same week that was then scored. Why 0.58 is not known. A wrong current-sensor ratio would give a constant weight like this, and so would a wye/delta error in the current path (0.58 is close to $1/\sqrt{3}$); a board only partly on this section would fit only if its on-section share were fixed, which is unlikely for a solar-and-plant board. None has been checked against the site's records. With every meter at face value the balance wobbles by ±30 kW and gives 17 false alarms. The "0" in the first row rests on that fitted weight, and the steady offset means the balance can show a change but not a constant island.
  - So, on the three-phase total, a 25 kW turn-on had no natural match in a week at five of eleven building mains and 1 to 9 at the rest. In the balance it had none only at the fitted weight, and the largest natural jump was 23 kW against a 25 kW threshold. It is one of about a hundred on a 5 MW head meter alone. Across single meters the count grows roughly with load and somewhat faster, with wide scatter among buildings (Figure 9). No turn-on was injected.
  - At 11 kW the three-phase total is a much weaker screen: no natural match at two of eleven building mains (25 and 96 kW), and the corrected balance has about five a day. The turn-on result on the total holds only at the 25 kW end of the range. The single-phase LBNL result above holds at both ends, since it had no natural step of 10 kVA or more.
  - Pauses: the corrected balance had no three-minute dip of 20 kW or more in the week, 0.7 a day of 15 kW or more, and 6 a day of 10 kW or more, the size of an 11 kW island's pause [M; 1-minute means, fitted weight].
  - A 5–7 MW feeder with a full balance is unmeasured here: we found no public data set with both the head meter and every customer meter at minute resolution or faster. Utilities hold such data.

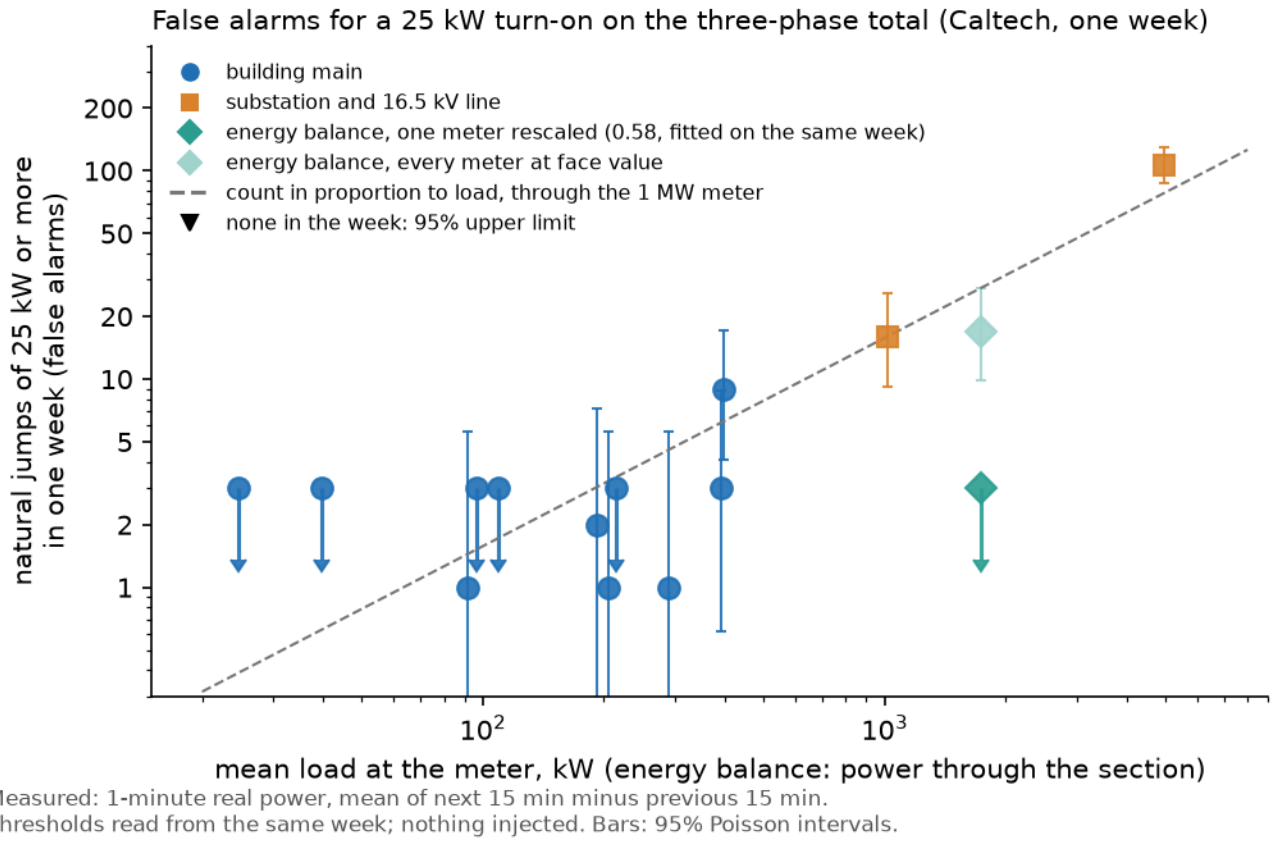


Figure 9. How often the grid produced a false alarm for a 25 kW turn-on in one week at Caltech, against aggregation level, on the three-phase total [M; thresholds read from the same week]. Zeros are drawn as upper limits: a count of 0 in one week bounds the rate below 3 a week at 95%. Bars on the other points are 95% Poisson intervals. The dashed line is a count proportional to load, drawn through the 1 MW substation meter, so that point agrees with it by construction. The largest meter sits above it (exponent about 1.2 between the two largest meters, 0.9–1.5 allowing for Poisson noise on the 1 MW count), and most mid-size building mains sit below it. The two node-balance points show how much the balance depends on rescaling one meter (weight 0.58, fitted on the same week). They are plotted at the power through the section, about 1.7 MW: the larger of the power in (two transformers and the inverter board, 1,737 kW) and the power out through the building mains (1,661 kW) [M].

- **Coincidence across meters, at Caltech.** Load steps on neighbouring building meters coincide about twice as often as independence predicts on the total (1,773 against 995, at 5 kVA or more, within ±2 s). On the phase unbalance, steps of 5 kVA or more are too rare to test (independence predicts 0.03 coincidences, and none occurred); at 2 kVA they coincide about 1.3 times as often as independence predicts (31 against 23) [M; seven building mains, one week].
- **Restart edges.** A synthetic 20 kVA island that pauses to exchange results (ideal square steps): 89% of restart edges recovered within ±2 s at a 15 kVA threshold fixed in advance, against 258 natural up-steps a day at that threshold on the three-phase total [S-syn; LBNL, 3.2 days]. So recovery alone does not separate a restart from the background; that needs periodicity or coincidence across islands. The sub-second edge counter cannot be tested on the logged job, because one-second means smear every edge.

2.5 Harmonics

Two kinds of power supply matter here. A plain six-pulse (three-phase bridge) rectifier draws each line's current in blocks rather than as a sine and emits strongly at the 5th, 7th, 11th and 13th harmonics (300, 420, 660 and 780 Hz). A supply with power-factor correction shapes its current toward a sine; its total distortion is a few percent at medium to full load (one 1 kW reference design: 1.4% at full load, about 2–7% at a third to a half of load), so each harmonic is a few percent at most. No injection was run; we measured what the buildings already emit at these frequencies and how much that varies from one capture to the next, and compared both with what a 25 kW island of each kind would add. The spread, not the total, is the yardstick, and at the 11th and 13th at the building main a corrected island is at most about twice it.

- **The meters can resolve harmonics.** One capture resolves a current harmonic to about 0.04–0.07 A at these meters [M; one ten-minute set per meter, 60 one-second captures].
- **The building already carries large ones.** At the building main the fifth harmonic is 13% of the 60 Hz current, with a standard deviation of about 3.4 A across the 60 captures [M].
- **A corrected island is lost at the low harmonics.** A 25 kW three-phase island with a corrected input adds about 0.6 A per harmonic at the building main [A-arith: assumes 2% per harmonic, a working figure from memory; no source found]. That is lost at the 5th and 7th. It is at most about twice the spread at the 11th and 13th, and less if, as is likely, a corrected input emits under 2% there [M spread, A-arith signal]. At 2.4 kV it is about equal to the spread, and at 16.5 kV below it. Harmonic current scales with the island's draw, so an 11 kW island (the logged job's own draw, section 0) adds under half as much: at most about the spread at the building main, and below it everywhere else [A-arith].
- **A plain rectifier would show.** A six-pulse rectifier's 11th and 13th would be about ten times the spread at the building main. Its 7th would be about ten times the spread at 2.4 kV, and about twice at 16.5 kV [A-arith on M].
- **Detection rates with false-alarm counts.** None, because no injection was run for harmonics.

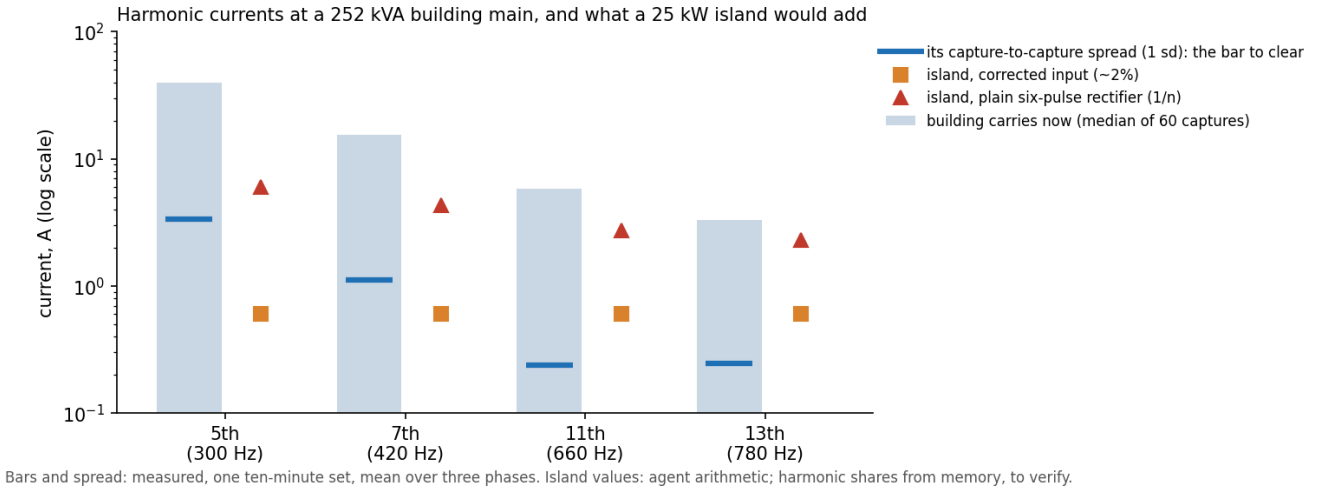


Figure 10. Harmonic currents at the 252 kVA building main [M], and what a 25 kW island would add [A-arith; corrected input assumed at 2% per harmonic, no source found]. The comparison that matters is the island against the capture-to-capture spread (blue dashes), not against the total. A corrected input is below the spread at the 5th and 7th and at most about twice it at the 11th and 13th. A plain rectifier clears it everywhere.

- **Limits.** One ten-minute set (60 one-second captures, one weekday at 11:00) at three meters. The building main is a research building whose large fifth harmonic suggests many drives (not confirmed). Harmonics add as vectors, so a new source can raise or lower the total.

2.6 The false-alarm finding

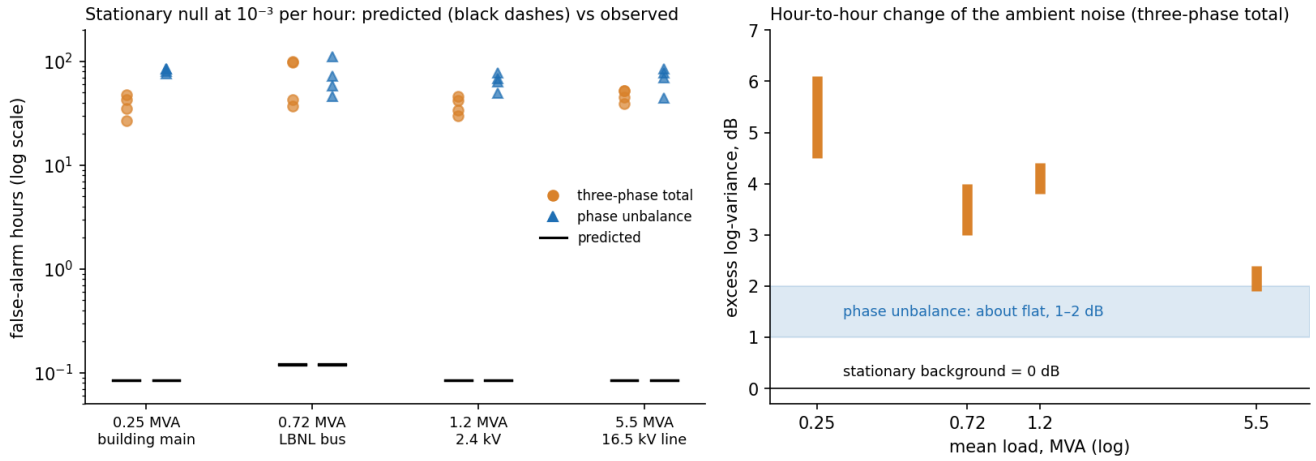
Every rate above depends on a threshold. The textbook threshold assumes a stationary grid background, and the background is not stationary; that is why the rates above use thresholds learned from each meter's own earlier hours. This section sets the textbook threshold's predicted false-alarm counts against the observed ones, says why they differ, and draws the consequence.

The measured failure is for the spectral statistic only. The background's hour-to-hour change probably affects the other detectors too, which is why each of them uses an empirical, per-meter threshold. Because false alarms are counted on no-signal hours, the finding does not depend on the pulse shape.

- **Result.** One test sets its threshold from a stationary chi-square null. This is the Follum–Pierre periodogram test (Follum & Pierre 2016), in a Welch-averaged form with an offline ambient spectrum from the learning hours. Aimed at 10^{-3} false alarms an hour, it predicts 0.12 false-alarm hours in 120 at LBNL and 0.084 in 84 at Caltech. On the test halves it gives 37–110 at LBNL and 27–84 at Caltech [M].
- **Cause.** The same thresholds fail on the hours their spectra came from: 40–95 at LBNL, 31–77 at Caltech [M, in-sample]. So look-ahead is not the main cause; in one LBNL row (phase unbalance, reverse direction) a shift between the halves roughly doubles the count, 110 against 57. The ambient noise changes from hour to hour. Its excess log-variance (the

variance of the log spectrum beyond what chi-square sampling alone gives) is 1.4–4.0 dB at LBNL and 0.9–6.1 dB at Caltech [M]. A stationary Gaussian background gives 0 dB by construction.

- **Trend with aggregation level.** On the three-phase total the non-stationarity is largest at the smallest meter: 4.5–6.1 dB at 0.25 MVA, 3.0–4.0 at 0.72, 3.8–4.4 at 1.2, 1.9–2.4 at 5.5 [M]. The two middle points come from different campuses and are out of order. On the phase unbalance it is roughly flat at about 1–3 dB. Four points, two campuses.
- **Correlated channels.** PNNL's correction for correlated channels (PNNL-24681) lowers the count by at most about 20 hours. It does not close the gap [M].
- **Consequence.** At the four meters tested, a stationary false-alarm model with an offline background estimate missed by two to three orders of magnitude, so the rates in section 2.3 use thresholds learned from each meter's own earlier hours. Even those failed at one meter and setting (74 false alarms in 84 hours at the building main, long-segment). Against the stationary prediction, the per-meter threshold detects about 8–14 dB higher at LBNL and 8–17 dB higher at Caltech, excluding the building main (4–23 dB across phases); most of this is the non-stationarity margin [M, read from the text; spot re-derived].
- **The empirical thresholds.** At margin 2, the stationary null gives them implied false-alarm probabilities from 10^{-66} to below 10^{-300} at LBNL, and 10^{-76} to below 10^{-300} at Caltech [M]. The heavy tails that break the null also warn against trusting our own threshold too far.
- **What is not claimed.** One statistic was tested. PNNL's online form estimates the ambient spectrum from the same data it tests, with a median filter; here that would mean the hour under test. It was not run. The supported claim is "the stationary null with an offline ambient estimate fails", not "forced-oscillation theory fails".
- **What better methods could do.** One statistic was tested, and a working rule from data science applies: try a plain method first; if it gets traction, keep digging; if it gets none, a better model will not save you. The plain method got traction on the single-phase case at every meter size (in the band chosen for each meter, margin 2 still finds every hour with at most one false alarm, and at LBNL 0.20–0.33 of the swing is enough, though about all of it at 5.5 MVA), so method work there is justified: a time-varying whitening model fitted before the spectral test, thresholds conditioned on hour of day, PNNL's online ambient estimate, and evidence accumulated across hours. At the logged swing it got none on the three-phase total at 0.7 MVA and above; an island 2 to 4 times larger is found at 0.7–1.2 MVA, about 11 times at 5.5 MVA. So there the next step is data more than method: per-phase and longer records (Part 4). Generic anomaly detectors would flag the same confounding events and are not the answer. [A-arith: reasoning, untested.]
- **What the zero counts mean.** No false alarm in 121 hours bounds the rate below about 18 a month (95%). In 84 hours, below about 26 a month; with one false alarm in 84 hours (5.5 MVA), below about 41 [A-arith]. These counts come from a band chosen per meter after scoring, which biases them low. Screening 1,000 to 10,000 lines at about one alarm a month in total needs about 10^{-3} to 10^{-4} alarms per line-month, four to five orders of magnitude below these bounds [A-arith].
- **An earlier instance.** This project's first thresholds were set on the 39 hours they were scored on. On 202 new hours they raised false alarms in 118, 121 and 41 hours [M]. Appendix B lists the figures this withdrew.



Measured on hours with no signal. Left: every band and fold, from thresholds_side_by_side*.json. Right: ranges from the results note, addenda 23-24. 0.72 MVA is LBNL; the rest Caltech.

Figure 11. Left: false-alarm hours that the stationary chi-square null predicts at 10^{-3} per hour (black dashes) against what each band and fold produced on no-signal hours [M]. The gap is two to three orders of magnitude at every level. Right: the hour-to-hour change in ambient noise on the three-phase total, by aggregation level [M, ranges from the results note]. A stationary background would sit at 0 dB.

2.7 Placement by signal

This table collects Parts 1 and 2 by where the island sits. Read down a column for what a monitor could see at a single-phase home or a three-phase business; across a row for how one signal changes between the two. The "nothing measured" cells are Part 4's open work.

Signal	Single-phase home or shop	Three-phase commercial premises
Load against expectation, at its own meter	About 20× an average US home's mean load [A-arith on A-chk: EIA]; a constant 12 kW load is found in every trial at single Spanish premises on hourly data [S-syn]	Ordinary in size (a restaurant averages about 24 kW) but flat around the clock; business-type comparison: nothing measured
Load against expectation, on a shared line	At chance at a 3.2 MW zone substation (energy mean shift, 15-minute data, 30 days) [S-syn]	Same
Pulse at the premises meter	Plain if the island is most of the premises' load, the rest is steady and the meter samples once a second [A-arith, not computed]; nothing measured	Untested at a premises; at a 252 kVA building main, 21% against 2 of 84 [S-job]
Pulse at a building main to a 5.5 MVA line	100% of hours, band chosen per meter after scoring; 1–4 false alarms in 84–121 at margin 1, 0–1 at margin 2 [S-job]	21% against 2 of 84 at 252 kVA (margin 1; 0% at margin 2); at chance at 0.7–5.5 MVA [S-job]
Turn-on step	No natural match of 10 kVA or more in 11.57 days on the phase unbalance at 724 kVA; at Caltech, no start or stop of 11 kVA or more in a week at 252 kVA and 1.2 MVA, none of 25 kVA or more at 5.5 MVA (8–13 a week of 11 kVA or more) [M]	0–9 a week at building mains; 0 in the energy balance only at a weight fitted on the same week (17 at face value); about 100 a week at 5 MW [M, one week]
Harmonics	Not modelled; rates: nothing measured	At the 252 kVA building main, a corrected input (2% per harmonic assumed) is lost at the 5th and 7th and at most about twice the spread at the 11th and 13th; about equal at 2.4 kV, lost at 16.5 kV; rates: nothing measured
Paper trail	27 kW is about 70% of a 200 A, 240 V service's continuous rating, leaving little room for household loads; an upgrade normally needs a permit, though unpermitted work is common [A-arith]	New premises: lease, business utility account, registration; existing premises (section 3.11) leave none [A-arith: judgment, untested]

2.8 Limits of these results

Eight limits, each with its remedy. Two concern the job (few workloads; counters, not wall power), one an unmodelled wiring case, three the grid records (two short campuses with balanced load; false alarms bounded, not measured; band setting chosen after scoring), one a loose fit, and one the meters themselves.

Each limit names what would remove it.

1. **Few workloads.** The single-phase result rests on one fine-tune (five near-identical runs, batch size 1 per GPU, ZeRO-3 CPU offload) whose step is a near-square 10-second pulse. Two single-node sessions from a second lab were run as checks: at Caltech a Llama-8B run reaches the same cells, and a diffusion run with a 16 s step and a smaller swing reaches them at margin 1 but not at 5.5 MVA at margin 2; at LBNL both are found in every hour on the phase unbalance (section 2.3). The first lab's Stable Diffusion run is found through blocks its source attributes to evaluation. No language-model pre-training job, no DiLoCo-style job and nothing faster than about 1 Hz is in the data. Needed: wall-side traces of several training styles.
2. **Software GPU readings, not wall power.** Needed: an AC-side trace of an 8–64 GPU host, per phase, at 10 samples a second or faster.
3. **Single-phase hosts in a three-phase service.** How a single-phase load shows on the primary depends on the service transformer. Behind a delta-wye transformer a 120 V line-to-neutral load appears on two primary lines at about 0.58 each in current (0.5 each in real power); a 208 V line-to-line load, the usual server wiring, appears on all three at 1/3, 2/3 and 1/3; behind a grounded wye-wye transformer the load stays on one line. Not modelled. On a delta-wye service we expect the threshold to rise about 1.7 times on current magnitude to about 2 times on real power [A-arith]. At the premises meter itself, a 208 V server load shows 0.58 of its swing on the phase unbalance, and how much of an island lands on one phase depends on how it is cabled. Section 3.1 gives the reach by installation, by arithmetic on stored results: full signal on residential service, about 0.58 for an improvised install (found to 1.2 MVA), 0.1–0.35 for a rack cabled per vendor guidance. Needed: a two- or three-phase split injected, not computed.
4. **Two campuses, 11.57 and 7 days.** Both backgrounds are mostly balanced three-phase load. That is why the phase unbalance is 3–13 times quieter than the total there [M]. On a residential feeder the phase unbalance is predicted to gain only about 1.4 times (one phase alone about 1.7 times), not about 8 [A-arith, assumes independent phases]. Single-phase hosts are mostly residential (US residential accounts outnumber commercial about 7 to 1) [A-chk: EIA 2024, Table 2.1; a proxy]. So the result is strongest where it has not been tested. Needed: per-phase data from a residential feeder.
5. **False-alarm rates are bounded, not measured** (section 2.6). Needed: longer records.
6. **Band setting chosen after scoring.** The two candidate band settings were fixed in advance (copied from LBNL), but the clean one differs by meter, even within Caltech, and which one to report was chosen after seeing the scored half. The LBNL test half has been reused across many analyses, so Caltech is only a partly clean repeat. The clean setting also depends on the job's step time (section 2.3), which a monitor does not know. Running both settings together, an hour alarming if either passes its own threshold, does not help: above 252 kVA it adds 0–3 false alarms in 84 hours and finds nothing the long setting misses, and the diffusion run's margin-2 result at 5.5 MVA stays at 18% / 58%, since the short setting finds it in no hour there. At 252 kVA the long setting alarms in 74 of 84 hours with no island (60 in the other direction), so that meter needs the short one [M on S-job; stored per-hour scores]. The choice matters at one meter, and there the no-island hours force it. Needed: a fresh record, with the band chosen per meter on early data alone.
7. **Ambient noise growth with load** is a loose fit: exponent 0.64 ± 0.15 to 0.78 ± 0.18 on the total, 0.45–0.60 on the phase unbalance depending on which meters are included (17 Caltech meters, one week; standard errors understated, since a meter's three phases are not independent) [M]. Buildings of one size differ 2–3 times.
8. **Meter error.** We have no measurement that splits meter error into fixed and moving parts; the one datum is the Caltech energy balance (Appendix A), from one node and one week, after one meter was rescaled. Offsets cancel in any before-and-after difference and a gain error only rescales a step; for a level comparison or a node balance, the fixed bias is what averaging cannot remove. The Caltech eGauge EG4X meters are certified to ANSI C12.20 class 0.5 (meter only); their instrument transformers add 0.5–1%, and more on lightly loaded circuits with oversized current transformers [A-chk: Caltech dataset paper, arXiv 2504.06588, Sec. IV.A]. Needed: a meter's drift measured against a reference over weeks.

Part 3. Evasion, and how we might get past it

Part 3 lists thirteen things an evader could do about the signals in Part 2, and what a monitor could do in return. Each counter has the same five slots: what it acts on, what it costs the evader, what signal remains, how a monitor might get past it, and what has been measured. The first and last slots show which counters matter and are untested. Three of the thirteen stand out:

balancing the load across three phases (3.1) costs nothing where three-phase service exists, takes only deliberate cabling, and removes both single-phase results; nesting inside a larger legitimate meter (3.11) is the one an evader would most likely use [A-arith: judgment]; fully off-grid generation (3.12) removes every grid-side signal and is outside this project.

General points.

- **The pulse result is dated to H100-era hardware.** Several counters arrive through general progress, not evasion: built-in power smoothing on newer chips, ramp and oscillation specifications some utilities impose on large training sites (Choukse et al. 2025; NERC's computational-load standards are in draft), and overlapped communication. None of these grid rules applies to a small island.
- **What no counter below removes entirely:** the island's energy has to come from somewhere (a grid meter, a host's meter, or fuel), it starts and stops, and its premises leave paper. Counters move these signals: off the grid (3.12), into a host's meter (3.11), or into a slow ramp (3.9).
- **The smoothing sources are easy to confuse.** The table below separates them. All rows are [A-chk] on Choukse et al. (2025, arXiv 2508.14318), read twice on this project and checked again on 2026-10-09, plus NVIDIA's GB300 blog (2025) for row 3. That reading awaits the author's ruling.

Mechanism	What the source says	What the source does not say
Built-in power smoothing (GB200 generation)	Floor at most 90% of rated power; 10.5% energy cost, simulated on a per-step waveform; limited lifetime	That it exists on H100s
Software filler ("Firefly")	Holds power up to rated; under 5% throughput cost, plus dedicated CPU cores and host bandwidth; energy overhead "similar" to the built-in feature's	An energy number or a leftover swing; no code is linked
Holding power through a stall	A programmable "stop delay" holds the GPU at its floor with no workload, then ramps down at the set rate (Fig. 5, GB200); NVIDIA's GB300 "power burner" holds constant power while it waits	Any duration limit or energy cost in numbers; availability on H100s

- **Each counter below gives:** what it acts on, its price, what it leaves, how a monitor might get past it, and status.

3.1 Balance the load across three phases

Part 2's two single-phase results, the pulse and the turn-on read on the phase unbalance, both depend on the island drawing all its power from one phase. An evader with three-phase service can split the load across phases at no cost, and the pulse must then be found on the three-phase total, where it is barely above chance at a building main and at chance from 0.7 MVA up.

- **Acts on.** Both single-phase results: the pulse and the turn-on on the phase unbalance.
- **Price.** Free where three-phase service exists, but not automatic. A server's power supplies share its load equally, so the split follows the cabling. Vendors advise rotating supply cords across phases, but the advice takes deliberate cabling, and NVIDIA's own tables leave one rack unbalanced ("it takes two racks of systems to balance the phases") [A-rep: vendor manuals, prior-art search of 2026-10-09]. Without three-phase service, the price is a lease or a service upgrade. Both leave paper.
- **How much of the signal is left, by installation.** The phase-unbalance read is linear, so the stored detection-versus-scale results give the share of the swing that must sit on one phase [A-arith on S-job: logged fine-tune, margin 1]. A 208 V server load sits across two lines and shows 0.58 of its swing on this read. Three tiers follow.
  - Residential 120/240 V service: the service transformer hangs on one primary phase, so the whole island is on one phase at the feeder however it is plugged in. Full signal; found at all four meters.
  - An improvised install on three-phase service (a shop or closet on one 208 V circuit or subpanel, or a rack corded A/B with no rotation): about 0.58. Found from 252 kVA to 1.2 MVA, lost at 5.5 MVA.
  - A rack cabled per vendor guidance: about 0.1–0.35, found at a building main at best; a deliberately balanced rack is the three-phase case measured above.

  The reach is for the fine-tune; the diffusion run of section 2.3 needs more swing at 1.2 and 5.5 MVA, so its tiers would reach less. No split was injected; this is arithmetic on stored results.
- **What remains.** The pulse on the three-phase total: above chance only at the 252 kVA building main, in time order at margin 1 (21% against 2 of 84); 0% at margin 2 and 10% against 5 in reverse; at chance at 0.7–5.5 MVA [S-job].

The turn-on step on the total low in the tree: no natural match at five of eleven building mains, and none in an energy balance only after one meter was rescaled on the same week; for an 11 kW island, two of eleven and 37 in the balance [M, false-match counts only; no turn-on injected].

- **How to get past it.** (a) A meter at the premises that samples once a second or faster. (b) An energy balance across meters low in the tree, for the turn-on and pauses. (c) Accumulating weak hours over months, which works only if the premises' confounding events come and go.
- **Status.** (b): its false-match background measured at one node for one week, after one meter was rescaled by a weight fitted on that week; no island signal injected. (a) and (c): nothing measured.

3.2 Software filler (a power floor in software)

Filler attacks the pulse directly. In the low phase of each step GPU power drops: on the logged job to about 79% of its peak (a 21–22% swing); in large jobs Choukse et al. report it can fall close to idle. Filler gives the GPUs junk work so power stays near its peak and the square wave flattens. The cost is energy. What survives is a short notch while the filler reacts, and the extra energy itself.

- **Acts on.** The per-step pulse. The filler keeps GPUs busy in the low-power phase.
- **Price.** Under 5% of throughput, plus dedicated CPU cores and host-device bandwidth for fast power polling [A-chk]. On the logged fine-tune it adds 10.9–11.7% to GPU energy, about 4% at the wall [A-arith]. This lands near the 10.5% in the table above by coincidence; the two measure different things. Engineering effort: not priced.
- **What remains.** A notch as long as the filler's reaction time. For a 10.2 s step it leaves 0.8% of the line at 25 ms, 3.1% at 100 ms and 31% at 1 s [A-arith bound]. Faster steps leave more. The extra energy remains.
- **How to get past it.** Sub-second notches of 15 kVA or more had no natural match on the phase unbalance at LBNL in 3.1 days (section 2.4). A filler notch is only as deep as the step swing (about 2.4 kW for the logged job); look-alike rates at that depth are not measured. A 120-sample-a-second phase-unbalance monitor is the candidate.
- **Status.** Not tested against a real filler. Extra energy as a signal: nothing measured.

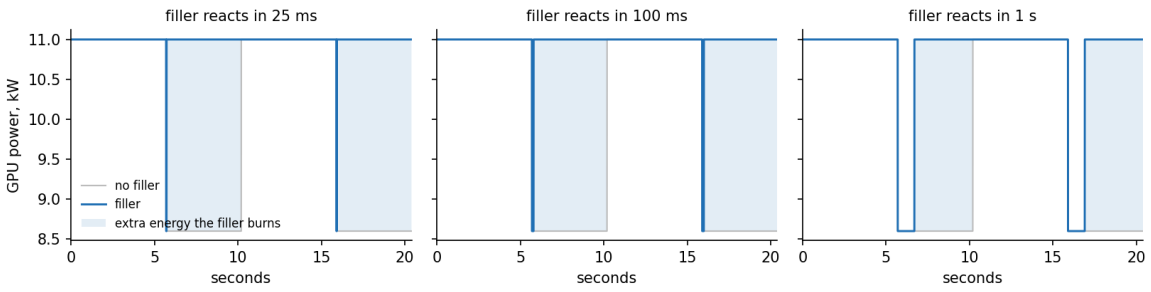


Figure 12. Illustration, not a measurement [A-arith]. A square step with the logged levels and timing. The filler holds GPU power at the high level once it reacts; what remains is one notch per step, as long as its reaction time. The shaded area is the extra energy it burns.

3.3 Built-in power smoothing (GB200 generation onward)

- **Acts on.** The pulse, the ramps, and power during a stall.
- **Price.** 10.5% energy at a 90% floor, simulated in the source. A limited lifetime [A-chk].
- **What remains.** The feature cannot hold fluctuations below 20% of rated power, from a 90% floor to a 1.1× short-peak ceiling [A-chk]. At one-second resolution the leftover swing is at most about 10% of rated power [A-arith]. During a stall it holds the floor for a programmable "stop delay", then ramps down at the set rate (Choukse et al. Fig. 5, GB200); the maximum delay and its energy cost are not stated.
- **How to get past it.** The leftover swing is a smaller pulse. Its detection: nothing measured. Ramps lasting about a second would also weaken the step's harmonics above about 0.5 Hz (section 2.3, Figure 8); the source does not state the ramp rates used.
- **Status.** Arrives by general progress. Not shown on H100s.

3.4 Battery, UPS or capacitors at the island

- **Acts on.** The pulse; on/off steps.
- **Price.** GB300's built-in store is 65 J per GPU [A-chk: NVIDIA GB300 blog, 2025]. That covers about 50 ms of full GB300 power (1.2–1.4 kW assumed, no source found) or about 0.4 s of the logged job's per-GPU swing of about 150 W; either way far short of a 10 s step [A-arith]. We found no sourced price for a seconds-duration, high-cycling battery at 1 MW or below.
- **What remains.** All the energy, plus conversion losses [A-arith]. A converter may add reactive power and harmonics, but a corrected front end keeps them small (section 2.5).
- **How to get past it.** Leads: household privacy studies of battery masking (simulated, one home) and unread patents on spotting behind-the-meter batteries in meter data [A-rep]. We found no peer-reviewed method, with error rates, that recognises a battery from upstream. A corrected converter input is lost at the 5th and 7th harmonics at a building main (section 2.5).
- **Status.** Mostly unmapped.

3.5 Burn through stalls with a junk load

- **Acts on.** The dip when a run crashes or stalls.
- **Price.** About a dollar a stall [A-arith].
- **What remains.** The gap between the crash and the junk load taking over.
- **How to get past it.** A notch of 15 kVA or more lasting 8–17 ms had 5–7 natural matches a day on one phase (1–3 at 20 kVA), none on the phase unbalance, and about 1,000–3,000 a day on the total at LBNL [M; one 0.7 MVA bus, 3.1 days]. A three-phase island's gap splits across phases, so for it only the total applies.
- **Status.** Crashes are rare for one island (section 1.3), though a covert island may fail more often. So this counter matters little.

3.6 Overlapped communication

- **Acts on.** Restart steps at each exchange, and their coincidence across islands.
- **Price.** At 100 Mbps, fully overlapped communication raises simulated compute use from 36% to 56% at 100 inner steps, at a small cost in model quality (perplexity 19.42 against 18.65) [A-chk: Koneputugodage & Long 2026, GlobalM1]. Streaming DiLoCo's own gains are simulated at 1 to 5 Gbit/s (arXiv 2501.18512); a reimplementation at 100 Mbps found no gain [A-chk: Koneputugodage & Long 2026, App. B.2].
- **Background.** In Koneputugodage & Long's simulator (1.5B model, 8 single-GPU workers), a blocking exchange at 100 Mbps lasts about 176 inner-step times: the compute share is 36%, 74% and 85% at 100, 500 and 1,000 inner steps [A-chk]. Their own wall-clock figures imply about 5 s per step, so roughly 10–15 minutes [A-src + A-arith]. Section 2.4 gives how well a blocking exchange shows.
- **What remains.** The island's own per-step pulse, which DiLoCo does not change (section 1.2). Whether anything shows in power at exchange time: nothing measured.
- **How to get past it.** Exchange timing seen from the network side. Nothing measured.

3.7 Staggered exchange times

- **Acts on.** Coincidence of restart steps across islands.
- **Price.** Nothing measured.
- **What remains.** Each island's own restart steps.
- **How to get past it.** Nothing measured. Natural coincidence on the total is already twice the independent rate at Caltech (section 2.4), so cross-island timing has a noisy baseline.

3.8 A cadence on a frequency the premises already has; deliberate jitter

The spectral detector looks for a line at the island's step frequency. If the premises already has a periodic load at that frequency, or the evader jitters the step time so the energy spreads across bins, the line is hidden. Both need knowledge of the premises or a throughput cost.

- **Acts on.** Spectral detection.
- **Price.** The evader needs knowledge of the premises. The throughput cost of jitter: nothing measured.

- **Why it could work.** The LBNL campus has periodic loads that come and go: 1.00 Hz, 3.17 Hz, 1.17 moving to 1.19 Hz, and 10 Hz [M]. The Caltech building main's long-segment setting is already at chance on the phase unbalance (section 2.3).
- **Counter-evidence.** Successive fine-tune steps are uncorrelated (lag-1 correlation −0.19 to 0.06) with a spread of 2.7–3.9% [M, weakly: log timing noise is about as large]. So natural jitter exists. Whether it costs detection: nothing measured.
- **How to get past it.** A per-premises choice of band setting. Notching known spectral lines.
- **Status.** Agent reasoning [A-arith]. Untested.

3.9 Slow start

- **Acts on.** The turn-on step, and the monitor's learning of the background.
- **Price.** Time.
- **What remains.** The running pattern. Three-minute dips of 20 kW or more had no natural match in a week in the Caltech energy balance (1-minute means, one meter at a weight fitted on the same week; 15 kW dips occurred 0.7 a day) (section 2.4). The pulse during training does not depend on how the island started. The longest window tested for a start is 60 minutes; a start ramped over longer than that is not a step by the section 2.4 rule.
- **How to get past it.** A frozen or certified baseline. Comparison with similar premises.
- **Status.** Nothing measured. Under the assumed threat model monitoring starts first, so every island has a turn-on inside the record. A slow start is the counter to that.

3.10 Falsely declared purpose

- **Acts on.** Load against what the premises should draw.
- **Price.** None.
- **How to get past it.** Audits of declarations; comparison by business type.
- **Context.** Grid reliability rules are moving toward declared use. Proposed changes to NERC's Rules of Procedure would register computational load sites of 50 MW or more at 100 kV or above, and draft standards under NERC Project 2026-02 would apply to them. Draft CLO-001-1 asks planners to collect each site's "expected use(s) or behavior(s)", with AI training or inference as a footnoted example [A-src for the drafts; the FERC order itself was not read]. That is far above island size.
- **Status.** Nothing measured.

3.11 Nesting inside a larger legitimate meter

Every meter-level result scales with the host: a 27 kW island behind a 250 kVA building is one problem, the same island behind a 5 MVA line another. This section extrapolates where the three-phase pulse would still show from the measured growth of ambient noise with load.

- **Acts on.** Every meter-level signal.
- **Price.** A willing or unaware host.
- **What remains.** The island's signals against the host's own variation.
- **Where the crossing sits.** The three-phase total finds the job in most hours at about 1.6 times its swing at the 252 kVA building main (time order, margin 2) [S-job]. With the measured growth of ambient noise with load, the logged swing alone would be found in most hours below about 120–140 kVA [A-arith]. The reverse direction at 252 kVA needs about 10 times the swing. So this figure carries a large uncertainty. A run at the 99 and 119 kVA building mains would measure it.
- **And on the phase unbalance.** The single-phase result has headroom of 3–5 times at 724 kVA (found in most hours at 0.20–0.33 of the swing, margin 2, time order). With the phase unbalance's noise growing as load to the power 0.45–0.60 (limit 7), the unscaled swing would stop being found in most hours at roughly 5–15 MVA [A-arith]. At the one point measured near there, the 5.5 MVA campus supply line, the headroom is already about 1×, and in reverse one band fails. So the reach of the single-phase result is about a large feeder, not beyond. A transmission line carries roughly 10 to a few hundred times the load at which the result runs out; the island is about 10^{-3} of a 115 kV line and under 10^{-4} only at 345 kV and above.
- **How to get past it.** Sub-metering. The turn-on test at Caltech found no natural 25 kW jump at five of eleven building mains, and no 11 kW jump at two (section 2.4).
- **Status.** Extrapolated, not measured.

3.12 Off-grid generation

- **Acts on.** Every grid-side signal, if fully islanded. Switching an existing load to a generator shows as a step down at the premises meter.
- **Price.** Fuel, permits, emissions, noise.
- **What remains.** Nothing on the grid.
- **How to get past it.** Fuel purchases, permits, thermal records.
- **Context.** On 10 July 2024 about 1,500 MW of data-center load switched to backup power during six faults in 82 s; about 1,260 MW stayed off the grid for hours (NERC incident review, January 2025) [A-src]. Grid accounting misses that energy. Backup engines at large data centers are often rated for unlimited running hours (Uptime Tier III and IV); what limits continuous running is their emergency status (US EPA: emergencies plus 100 hours a year) and on-site fuel [A-chk: Cummins 2017; 40 CFR 60.4211(f); fuel limit no source found].
- **Status.** Outside this project's measurements.

3.13 A faked reactive component, to look like a motor

- **Acts on.** The test that separates steps by their real and reactive parts.
- **Price.** Not priced.
- **What remains.** Timing between the real and reactive edges.
- **How to get past it.** Natural step statistics, including steps with no reactive part, are in section 2.4; motor starts are in Figure 4 and Appendix A. The timing between natural real and reactive edges is measured in the project's results note but not reported here.
- **Status.** No faked step tested.

Part 4. Open cells and asks

1. **Wall-side traces of real training** on 8–64 GPUs, at 10 samples a second or faster, per phase if possible, with step and exchange times logged. More than one training style, and at least one overlapped-communication run. Removes limits 1 and 2.
2. **Longer and more varied line data.** The rest of the LBNL record (its citation names three months; the public file holds 11.57 days). Any residential feeder with per-phase data faster than one sample a second. A feeder with its head meter and every customer meter. Removes limits 4–6.
3. **A per-phase measurement at a GPU host.**
4. **Method.** Is there an established way to set thresholds in non-stationary ambient noise that we should have used first? Section 2.6 lists our candidates. Does a measured feeder-head power spectrum for 0.01–60 Hz exist? We did not find one in a limited search.
5. **For Rahman.** What "monitoring threshold" was meant to cover. What step time and exchange schedule he considers realistic for one node.
6. **Cheap runs on data in hand.** The small-host crossing at 99 and 119 kVA (section 3.11). An injection for harmonics (section 2.5). A start and stop injected on the Caltech phase unbalance, including a start ramped over more than an hour (section 2.4). The band setting chosen from no-island hours of the learning half alone (limit 6).

Appendix A. Grid background

Figure 1 shows each level. The table gives typical US values. **Cells checked on 2026-10-09 come from Short's distribution handbook (via electrical-engineering-portal.com and Iowa State EE455 slides), the DOE 2015 electricity primer and surge-impedance-loading tables [A-chk]. Transmission and subtransmission line lengths, subtransmission power and panel circuit counts are agent recall [A-arith; no source found].** These are not yet checked against a primary source.

Level	Kind	Length (wires) or exits (nodes)	Voltage, power
Transmission line	wire	tens to hundreds of km	69 or 115–765 kV; tens of MW (115–138 kV) to about 2 GW (765 kV)

Level	Kind	Length (wires) or exits (nodes)	Voltage, power
Transmission substation	splitter and transformer	several lines in and out (a mesh, not a tree)	steps down to subtransmission, 34–69 kV (115–138 kV at some utilities)
Subtransmission line	wire	length not sourced	34–69 kV; power not sourced
Distribution substation transformer	transformer	1–2 or more lines in; 1–6 transformers, typically 5–60 MVA each	steps down to 4–35 kV; about 5 MVA (rural) to over 200 MVA (urban) per station
Substation bus and feeder breakers	splitter	typically 4 feeders per bus (1–8), about 8 at a two-transformer station	4–35 kV
Feeder mainline	wire	3–24 km (2–15 mi), up to about 50 km at 34.5 kV	1–15 MVA peak, typically 7 MVA
Taps along the feeder	splitter	1 in, 2 out at each; about 15–30 single-phase laterals on a residential feeder	same voltage
Lateral	wire	0.3–8 km (0.2–5 mi)	up to several hundred kVA on one phase
Service transformer	transformer	1 in; 1 secondary out to one or a few homes (US), 100–300 customers (Europe)	to 120/240 V, 120/208 V or 277/480 V (US), 230/400 V (Europe); 10 kVA to about 2 MVA
Secondary and service drop	wire	under about 75 m (US); up to about 1.6 km (Europe)	120/240 V (US); 230/400 V (Europe)
Meter and panel	splitter	1 in; a few dozen circuits out	—

- A feeder serves typically a few hundred to a few thousand customers (most common 400, range 50–5,000) through roughly 50–450 service transformers [A-chk: Short; PNNL-18035]. Mainlines are three-phase, usually four-wire with a grounded neutral; laterals are single-, two- or three-phase; residential services are single-phase.
- Per-feeder monitoring means roughly 4–12 three-phase monitors per substation. Monitoring service transformers multiplies that by hundreds [A-arith].
- **Where meters sit.** Most feeder heads have a digital relay or a recording meter, though some substations still have only analog ammeters; some mid-feeder reclosers meter; every customer has a revenue meter.
- **Feeder size.** A typical feeder is 400 A, about 7 MVA at peak across three phases, in a range of 1–15 MVA [A-chk: Short's table via electrical-engineering-portal.com; Iowa State EE455 slides]. Feeders of about 17–22 MVA exist at 23–35 kV (PNNL-18035; Florida PSC 2016). One phase carries about 2.3 MVA at peak.
- **The island as a share.** About 0.4% of a typical feeder at peak, about 1% of one phase at peak [A-arith].
- **Phases.** In North America larger commercial services are usually three-phase, and a home's transformer takes one primary phase (two on three-wire primaries); in Europe many homes have three-phase service. How a single-phase load inside a three-phase service shows on the primary depends on the service transformer (limit 3).
- **Loads.** A residential water heater is single-phase and switches in one step per element; commercial ones are often three-phase but still switch one element at a time. Commercial air handlers are usually three-phase. A motor started directly across the line draws about 5–8 times its running current for a moment (NEMA Design B up to about 6.4 times) [A-chk: SIU motor-starting notes]; one clean start on the LBNL bus read 3.7–3.9 times in current magnitude, which understates it [M, one event]. Motors on soft starters or drives draw little or no inrush.
- **Real and reactive power.** Real power does work. Reactive power moves back and forth with motors and some converters. The phase angle between voltage and current sets the split. A step with almost no reactive part is unlike a motor started across the line, but looks like a motor behind a variable-speed drive (section 2.4).
- **Phase unbalance.** Here, one phase minus the mean of the other two, on current magnitudes. It has no exact standard name; the closest is "current unbalance", the largest deviation from the mean magnitude. The IEC measure is negative-sequence over positive-sequence current, a related phasor quantity. Power-quality meters and many relays compute both [A-src: SEL-751, SEL-735 and Chauvin Arnoux texts; the standards are paywalled].
- **Meter classes and what moves.** Revenue (billing) meters are certified to ANSI C12.20 class 0.5 or 0.2; ordinary meters fall under ANSI C12.1, about ±2% in service; relay metering at the feeder head is about ±1–3% on real power before current-transformer error [A-chk: meter-accuracy check of 2026-10-09; C12.20 via a UL proxy]. A class bounds the meter's own deviation over a range of load and power factor, and excludes instrument-transformer error. Our one measurement of what moves is the Caltech energy balance: after one meter is rescaled by a weight fitted on the same week, the balance of about 14 meters changes by about 5 kW from one minute to the next on 3.4 MW of metered flow

(0.14%); its hourly means spread by about 9 kW (0.26%) and its residual by ±11 kW (0.3%) [M; one node, one week]. We have no measurement that splits meter error into fixed and moving parts. For a change (turn-on, pause) offsets cancel and a gain error only rescales the step, so steadiness matters. For a level comparison (between meters, or against a declared load) the whole measurement chain is the limit, and the fixed bias is what averaging cannot remove: 25 kW is one error band at 0.5% of 5 MW and two and a half at 0.2% (an 11 kW island is under half a band at 0.5% and about one at 0.2%), and with relay metering at 1–5% a level comparison cannot see one island.

- **What is installed, and what it would take to use it.** Nearly every feeder leaving a distribution substation has current transformers feeding protection; most also have voltage transformers, and many report to SCADA [A-chk for relays; SCADA coverage from search snippets only]. 60% of US distribution feeder relays were digital in 2012; in the 2009 survey, 97% of planned purchases were digital [A-chk: Newton-Evans]. A digital feeder relay samples the phase currents, and usually the voltages, at 32 samples a cycle or more (about 2 kHz; one recloser control 128 a cycle); some models are current-only [A-chk: SEL-751, SEL-651R-2 and ABB REF615 data sheets]. So per-phase current is already sensed at most feeder heads, though at relay-metering accuracy (about 1–5% on real power) and with unmeasured resolution for a 27 kW step, which is about 10–30 mA on the current transformer's secondary, below the SEL-751's specified-accuracy floor. About 40% of feeder relays were still electromechanical in 2012. What a relay does not do today is keep its samples: it stores a few seconds of waveform when a fault triggers it and reports averaged metering to SCADA every few seconds. So the gap is recording, reporting and metering accuracy, not only sensing. The section 2.3 pulse tables used one-second means of each phase, averaged from a faster stream (the other five LBNL runs used a 12-a-second series). A monitor would need to stream several readings a second and average them; a single unaveraged reading each second would fold the site's own 1–3 Hz loads into the search band. That is within synchrophasor streaming (IEEE C37.118, 1–60 messages a second), which many current relays support, though it needs a time source, a data concentrator and communications [A-chk: SEL-751 data sheet; the standard is paywalled]. The notch results were measured at 120 a second with the µPMU used here; notches of 8–25 ms need that rate, and longer ones were not tested at lower rates. No count exists of relays that can, or are set up to, stream. A distribution PMU costs on the order of several thousand dollars as a device (NASPI 2018), but installation and communications can cost more, and no sourced installed cost per feeder exists.
- **Periodic swings travel far.** On 11 January 2019, a Florida steam turbine swung about 200 MW peak-to-peak at 0.25 Hz for about 18 minutes. It showed as about 50 MW in New England (NERC, Eastern Interconnection Oscillation Disturbance, December 2019) [A-chk].
- **Large loads drop together.** On 10 July 2024 about 1,500 MW of data-center load moved to backup power during a sequence of six cleared faults in 82 s, most of it at the third; about 1,260 MW stayed off for hours (NERC incident review, January 2025) [A-src]. PJM reports about 3,800 MW on 22 July 2026 in two waves (PJM Operating Committee, August 2026) [A-src].

Appendix B. Withdrawn figures

Figure	Where it appeared	Why withdrawn
6.8, 14.6, 6.2–6.6 and 1.8 kVA for detection in 80% of hours, synthetic pulse	Early results	Thresholds set on the hours scored. On 202 new hours they raised 118, 121 and 41 false alarms. Also read off a grid too coarse by about ±30%
"Found in 120 of 120 hours", 1.0–1.2 kW, about 32 kW on the total	Logged-job injection, first version	Learned on the later half (look-ahead). Replaced by section 2.3
Spectral 2.7–3.0 kVA (phase unbalance), 18.5 kVA (total)	Synthetic pulse	Look-ahead; in time order the full-band search raised 37 and 34 false alarms in 121
A 13-second step for the fine-tune	Secondary reports	The step is 10.1–10.3 s. 12.8–13.0 s is Stable Diffusion's non-training block cycle
Implied false-alarm probability of 10^{-26}	One reading of section 2.6	That threshold was margin 1, which did raise false alarms
The frontier table in the original results note	Early results	Superseded twice. It rests on assumed scalings and a synthetic signal

Appendix C. Citations to verify

What is still unverified after the 2026-10-09 source check of 462 claims: claims tagged [A-rep] or marked [no source found] above, and sources cited at second hand. None has yet been checked against its primary source.

- The Epoch/SemiAnalysis figure of about 1.4 kW IT and 1.7 kW at the grid per H100, which carries the 27 kW (not in the repo).
- Epoch AI 2026b, cited by Rahman for tracking large clusters by power and footprint.
- Grid modelling cadences of 0.5–1.5 Hz (Ko & Zhu 2026, resting on the NERC Large Load Task Force 2025 report, not read); the NASPI/PNNL webinar figure of 0.1–1 Hz, not located.
- Supraharmonic (2–150 kHz) emissions of server supplies: no measurement found.
- Full GB300 GPU power (1.2–1.4 kW), used to turn 65 J into 50 ms.
- NVIDIA's guidance on spreading power supplies across phases.
- Behind-the-meter battery patents (US 12,562,572; US 2024/0258796), not read.
- Most US premises meters recording every 15 or 60 minutes (utility examples only); feeder SCADA coverage (search snippets only).
- "No published feeder-head spectrum for 0.01–60 Hz": a limited search found none (IEEE Xplore not covered).
- The FERC order on NERC's computational-load registration (RD26-7-000, 16 July 2026).
- Paywalled standards behind vendor or proxy text: ANSI C12.20 (UL proxy), IEC 61000-4-30, IEEE C37.118, the NEMA current-unbalance definition, and the NEC 80% continuous-load rule.
- The Caltech egauge_17 and egauge_19 week files (1,160 and 5,489 kVA traced to stored results, not recomputed from raw data).
- Not a report, but unsourced: the 2% per-harmonic share of a corrected input and the six-pulse shares (section 2.5); an installed per-feeder monitor cost; subtransmission line lengths and power, transmission line lengths and panel circuit counts (Appendix A); on-site fuel as the limit on backup running (section 3.12).

Appendix D. Detection tables, both directions

The tables of section 2.3 with both band settings and the reverse direction added. Reverse learns the threshold on the later half of the record and scores the earlier half, so it has seen the future; it is shown as a check on how much the result depends on which half the threshold came from. A detection share means little where a third or more of no-signal hours alarm; those cells are marked "at chance".

Single-phase island (phase unbalance). Logged fine-tune, first run, unscaled [S-job].

Aggregation level	Setting	Time order, margin 1	Time order, margin 2	Reverse, margin 1	Reverse, margin 2
Caltech building main, 252 kVA	short-segment	100% (1/84)	100% (0/84)	100% (13/84)	100% (1/84)
	long-segment	100% (74/84), at chance	100% (74/84), at chance	100% (60/84), at chance	100% (6/84)
LBNL 12 kV bus, 724 kVA	short-segment	100% (1/121)	100% (0/121)	100% (4/120)	100% (0/120)
	long-segment	100% (2/121)	100% (0/121)	100% (1/120)	100% (0/120)
Caltech 2.4 kV section, 1,160 kVA	short-segment	100% (2/84)	100% (0/84)	100% (0/84)	100% (0/84)
	long-segment	100% (4/84)	100% (0/84)	100% (0/84)	100% (0/84)
Caltech 16.5 kV line, 5,489 kVA	short-segment	100% (3/84)	45–95% (0/84)	0–6% (3/84)	0% (0/84)
	long-segment	100% (4/84)	100% (1/84)	100% (6/84)	100% (0/84)

Three-phase island (three-phase total). Same job and design [S-job].

Aggregation level	Setting	Time order, margin 1	Time order, margin 2	Reverse, margin 1	Reverse, margin 2
Caltech building main, 252 kVA	short-segment	5% (3/84)	0% (0/84)	1% (0/84)	0% (0/84)
	long-segment	21% (2/84)	0% (0/84)	10% (5/84)	0% (0/84)
LBNL 12 kV bus, 724 kVA	short-segment	4% (3/121)	0% (0/121)	1% (1/120)	0% (0/120)
	long-segment	6% (3/121)	0% (0/121)	4% (4/120)	1% (1/120)
Caltech 2.4 kV section, 1,160 kVA	short-segment	2% (2/84)	2% (2/84)	1% (1/84)	0% (0/84)
	long-segment	5% (4/84)	0% (0/84)	2% (2/84)	0% (0/84)
Caltech 16.5 kV line, 5,489 kVA	short-segment	2% (2/84)	0% (0/84)	0% (0/84)	0% (0/84)
	long-segment	2% (2/84)	0% (0/84)	2% (2/84)	0% (0/84)

Aggregation level	Setting	Time order, margin 1	Time order, margin 2	Reverse, margin 1	Reverse, margin 2

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